Introduction to Computer Graphics

Tao Du

Tsinghua University & Shanghai Qi Zhi Institute

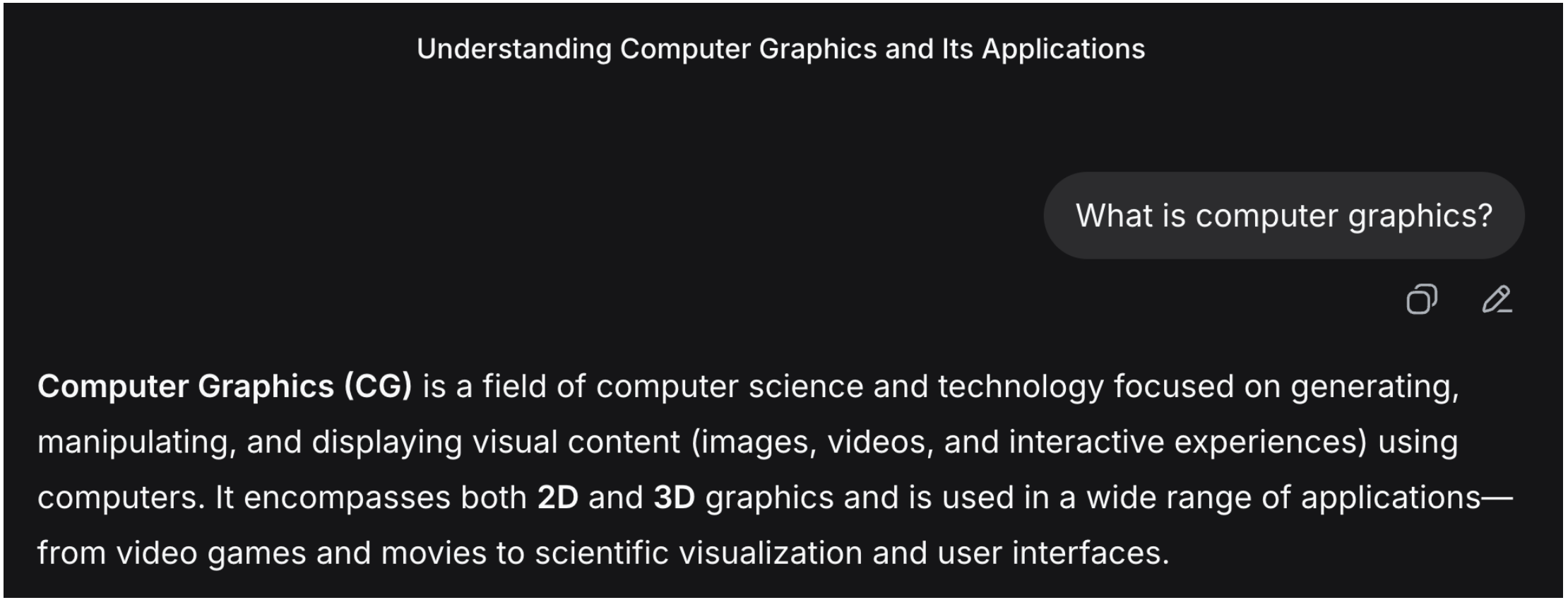
Talk outline

The foundation: building blocks of a virtual world

The next horizon: research frontiers of graphics

What is computer graphics?

An answer from DeepSeek:



The early days of graphics: Sketchpad (1963)

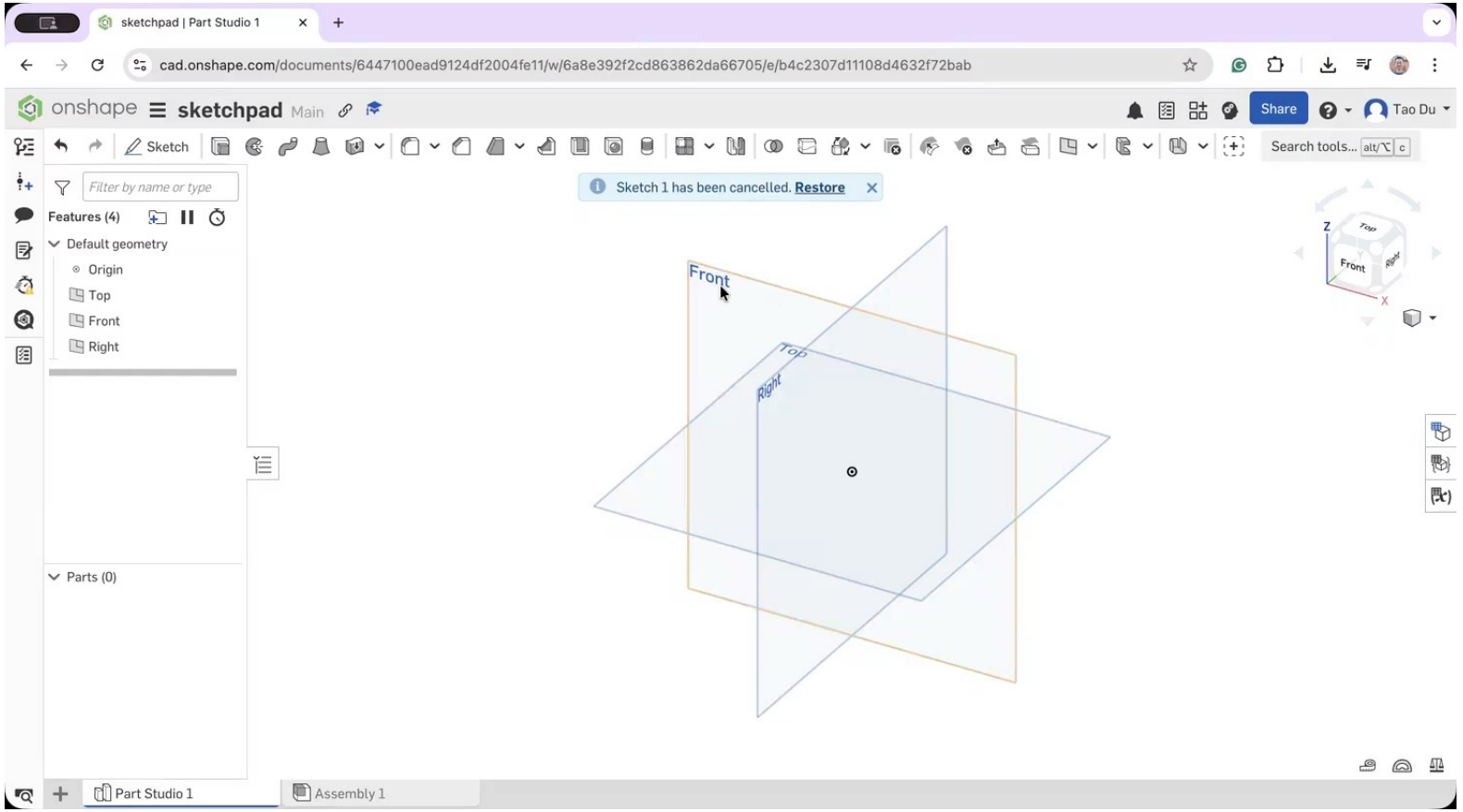
Sketchpad is a computer program invented by Ivan Sutherland (ACM Turing award winner in 1988)



Source: https://www.computerhistory.org/collections/catalog/102706673/?media=138328

The early days of graphics: Sketchpad (1963)

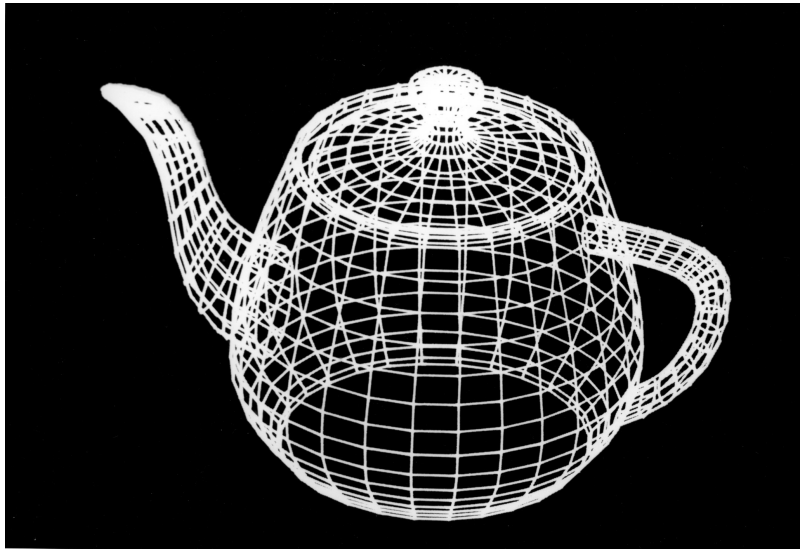
Modern CAD systems inherit many of the key designs in Sketchpad, e.g., geometric constraints.

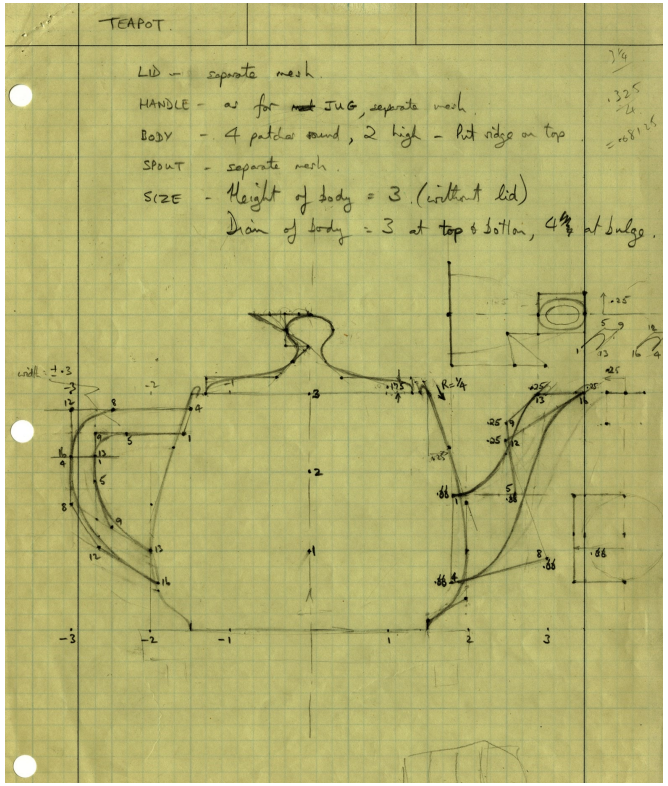


The early days of graphics: Utah teapot (1975)

The “Hello, world!” moment of 3D modeling, created by Martin Newell.





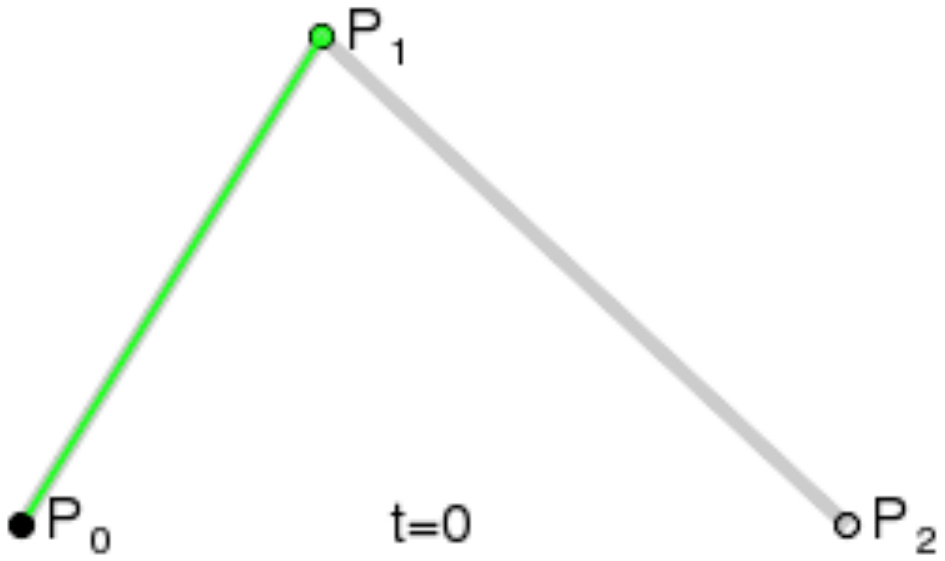


Source: https://www.guidigo.com/Web/CHM-Revolution-Tour/DHTH2i2fYjA/Stop/18/Computer-Graphics-Music--Art-Utah-Teapot

The early days of graphics: Utah teapot (1975)

“The teapot was one of the first to be modeled using Bézier curves.” (Wikipedia).

Can you give an equation describing the red curve?



Source: https://en.wikipedia.org/wiki/B%C3%A9zier_curve

The early days of graphics: Utah teapot (1975)

Utah teapot in today's graphics research:

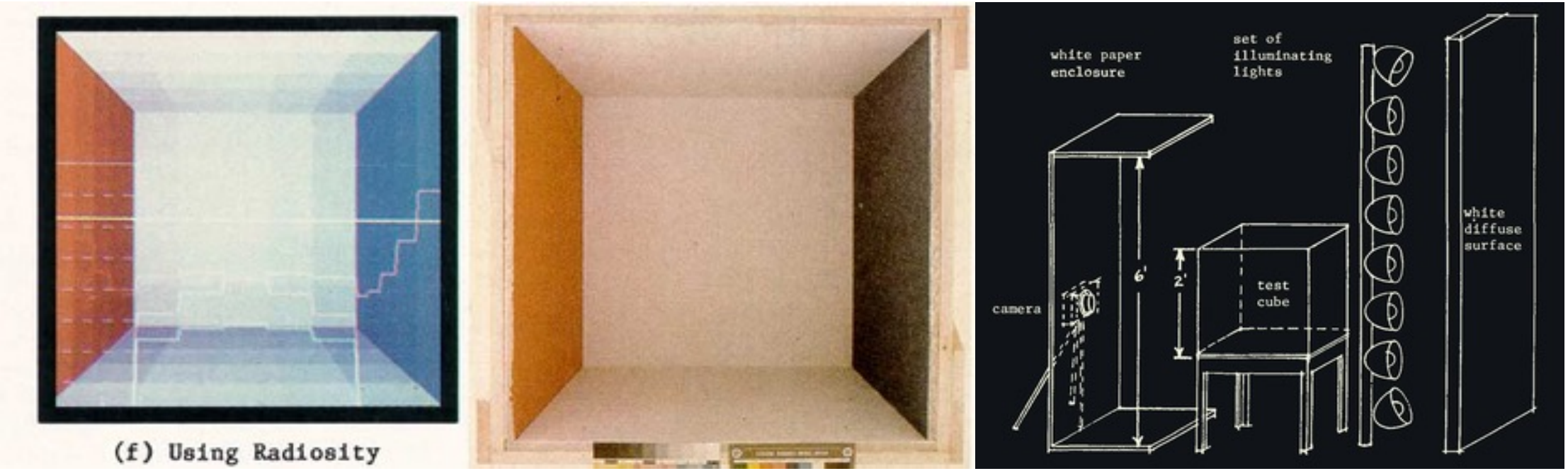




Source: research papers from Prof. Yin Yang (left) and Dr. Kui Wu (right).

The early days of graphics: Cornell box (1984)

In 1984, Cindy Goral at Cornell created this famous physical environment for rendering research.



Source: https://en.wikipedia.org/wiki/Cornell_box

The early days of graphics: Cornell box (1991)

Researchers at Cornell introduced a new version in 1991, which became the Cornell box we are familiar with today:

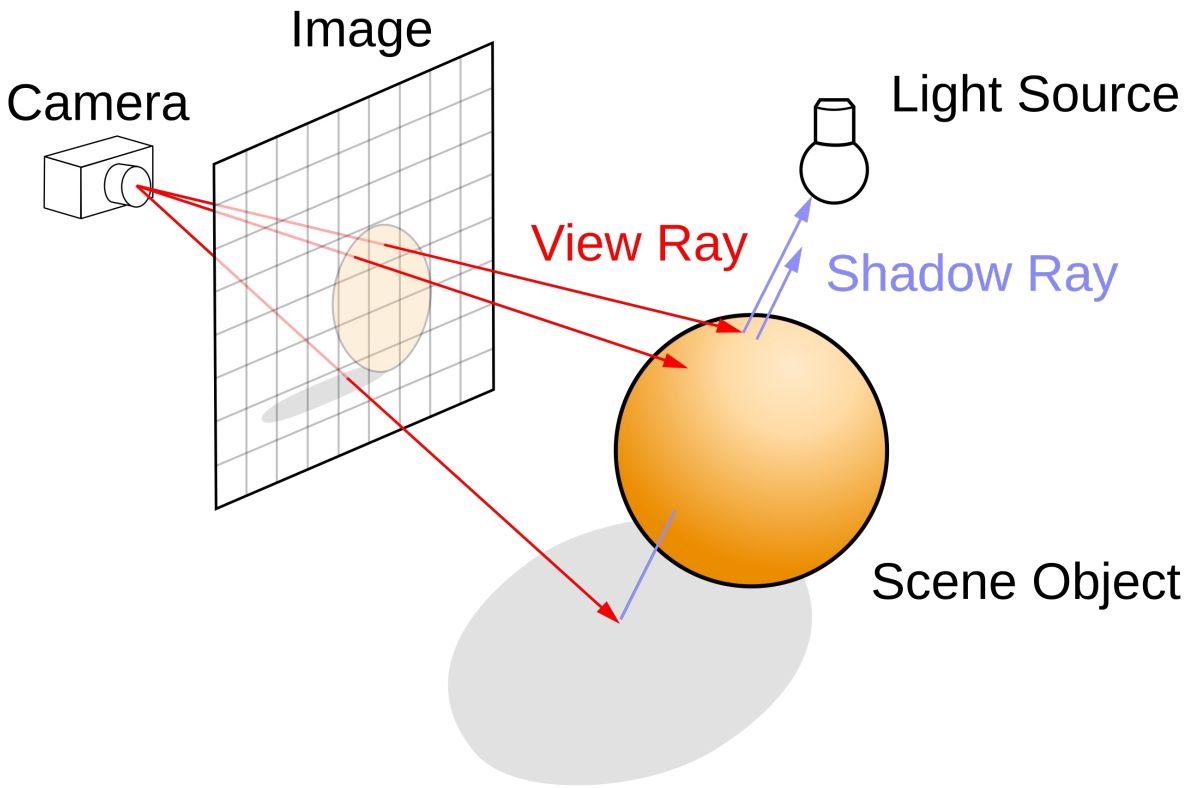




Source: https://bowers.cornell.edu/cornell-box

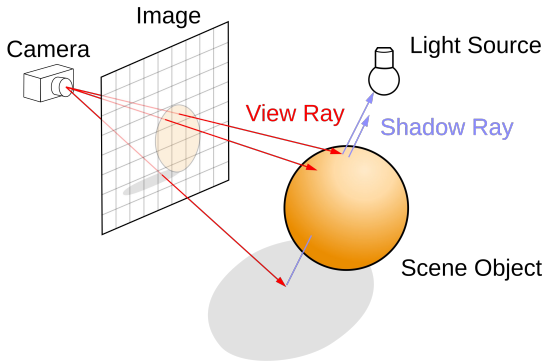
The early days of graphics: Cornell box (1991)

An overview of the ray tracing pipeline:



Source: https://en.wikipedia.org/wiki/Ray_tracing_(graphics)

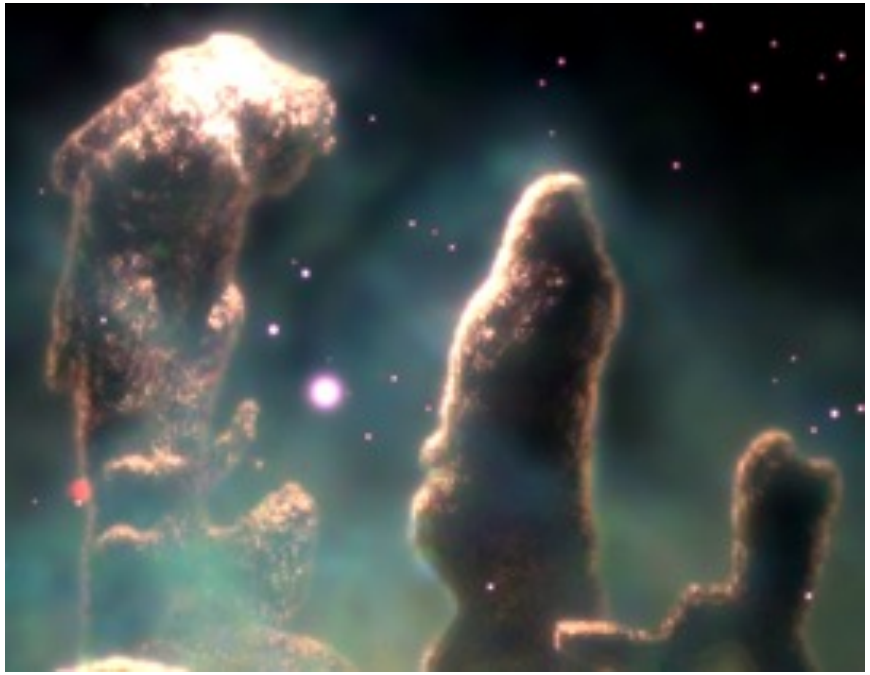
The early days of graphics: Cornell box (1991)



CS348B rendering competition at Stanford (1992-2019?)



Huamin Wang
(2003, first prize)



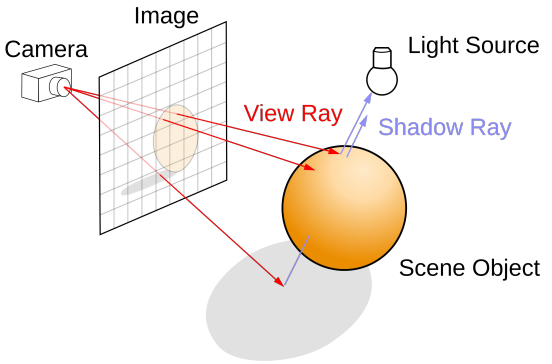
Sergey Levine and Edward Luong
(2008, grand prize)



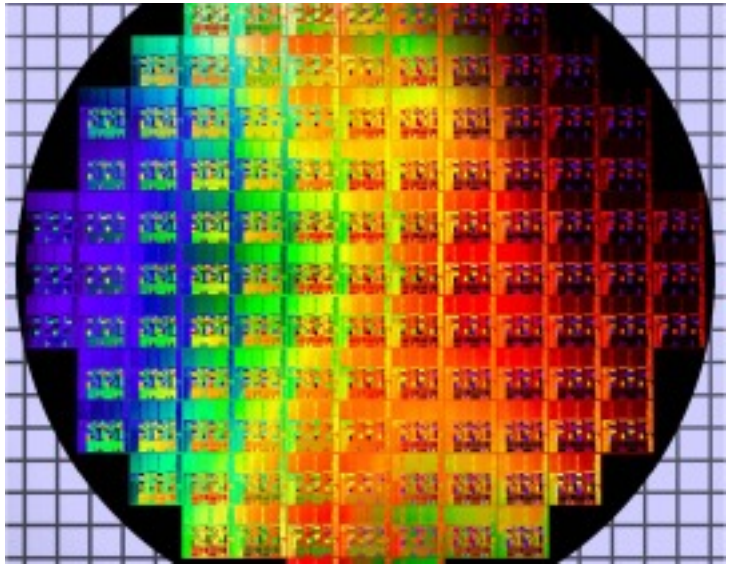
Ben Mildenhall
(2013, grand prize)

Source: http://www.graphics.stanford.edu/courses/cs348b-competition/

The early days of graphics: Cornell box (1991)



CS348B rendering competition at Stanford (1992-2019?)



Mingyu Gao and Jing Pu
(2013, first prize)



Saket Pankar and Bo Zhu
(2014, grand prize)

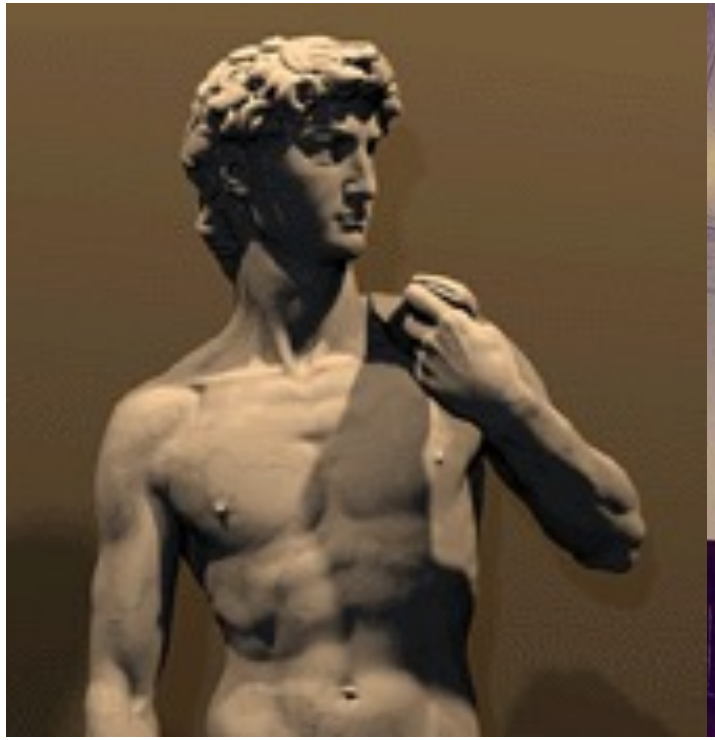


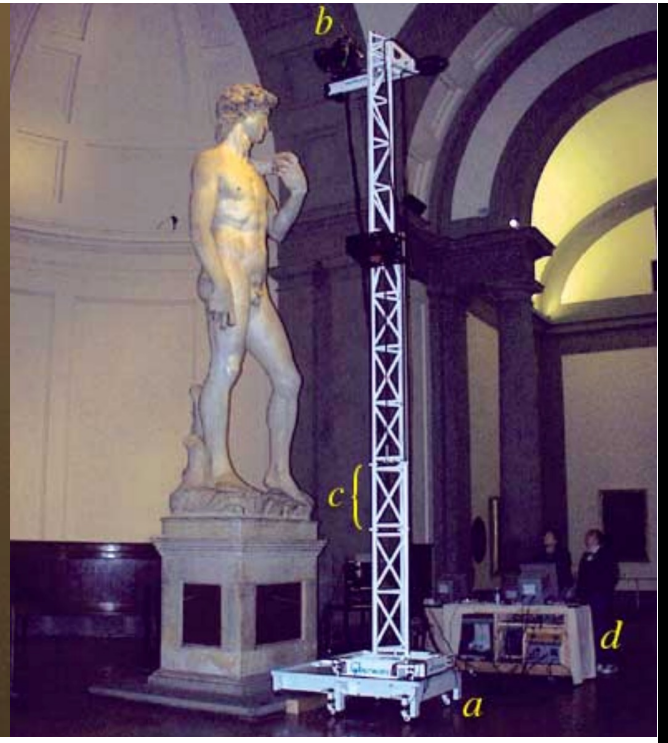
Jennifer Tao and Cynthia Jia
(2019, grand prize)

Source: http://www.graphics.stanford.edu/courses/cs348b-competition/

The early days of graphics: digital Michelangelo (2000)

A project led by Stanford scanned Michelangelo's David to build a 3D model with 8 million polygons.







Source: https://graphics.stanford.edu/papers/dmich-sig00/

The early days of graphics: digital Michelangelo (2000)

Label the two pictures with photograph and rendering.

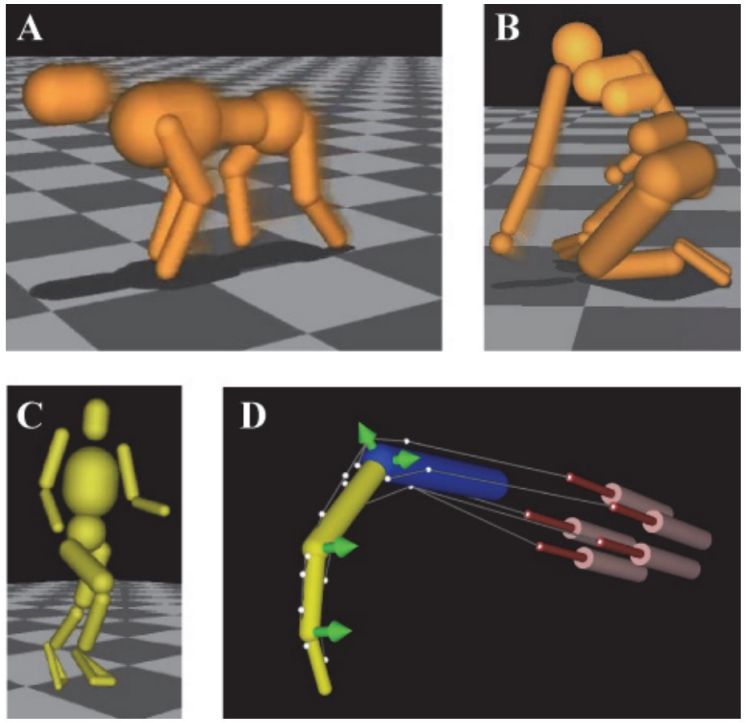




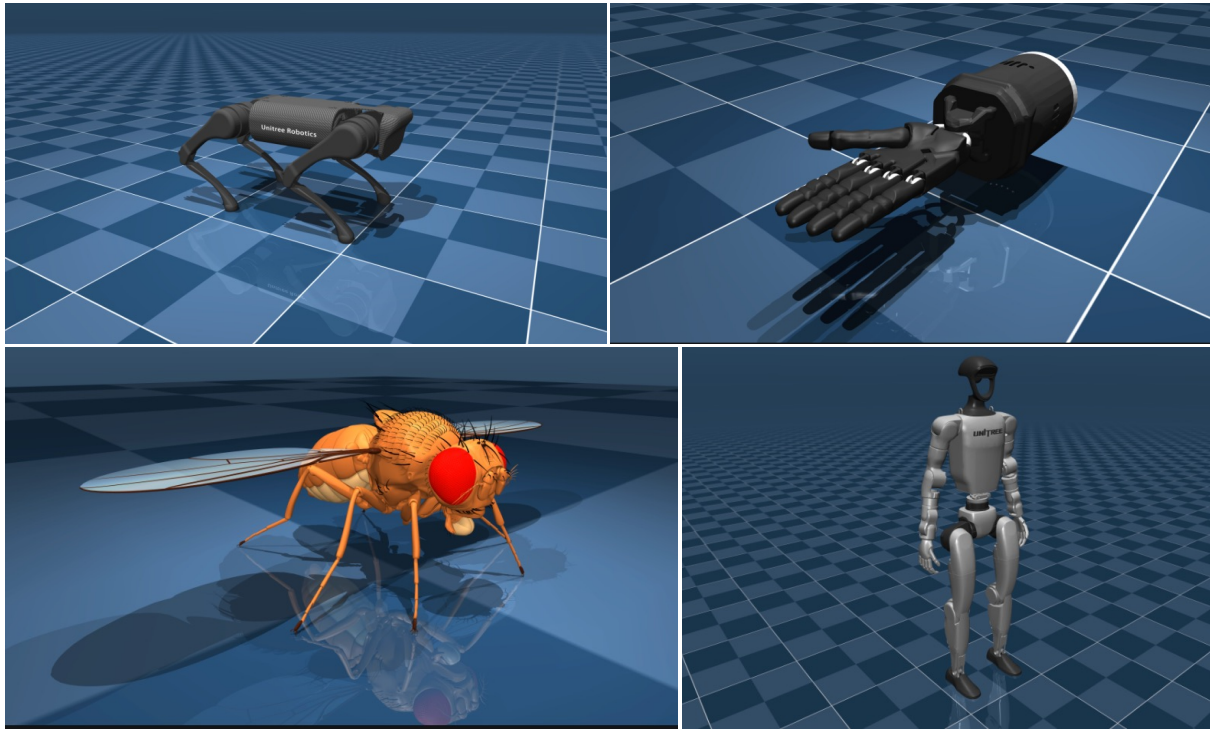
Source: https://graphics.stanford.edu/papers/dmich-sig00/

The early days of graphics: physics animation (2010s)

As graphics hardware became increasingly powerful, researchers started to develop physics engines that later turned into influential research platforms today.



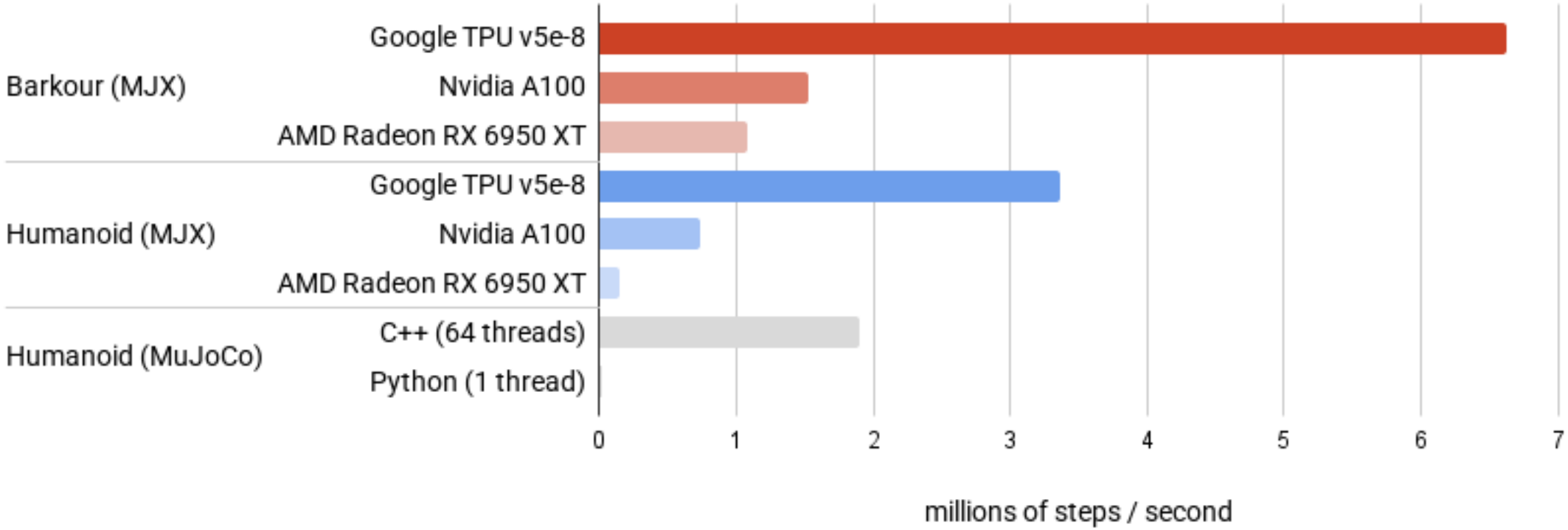
MuJoCo (2012)



MuJoCo (2025)

The early days of graphics: physics animation (2010s)

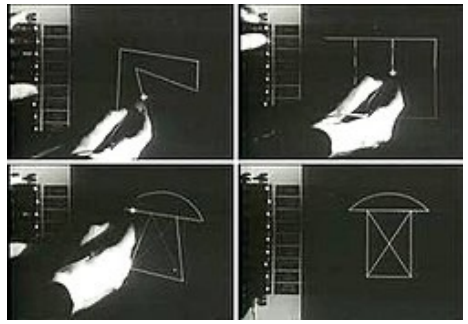
Rigid-body robot simulators can be extremely fast today.



Source: https://github.com/google-deepmind/mujoco/discussions/1101

The early days of graphics: summary

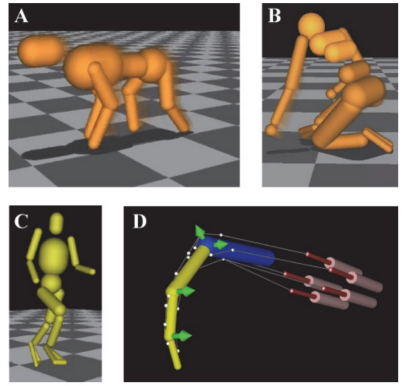
User interface



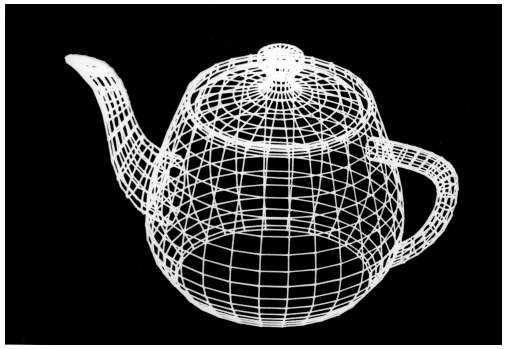
Rendering



Animation



1960s 1970s 1980s 1990s 2000s 2010s



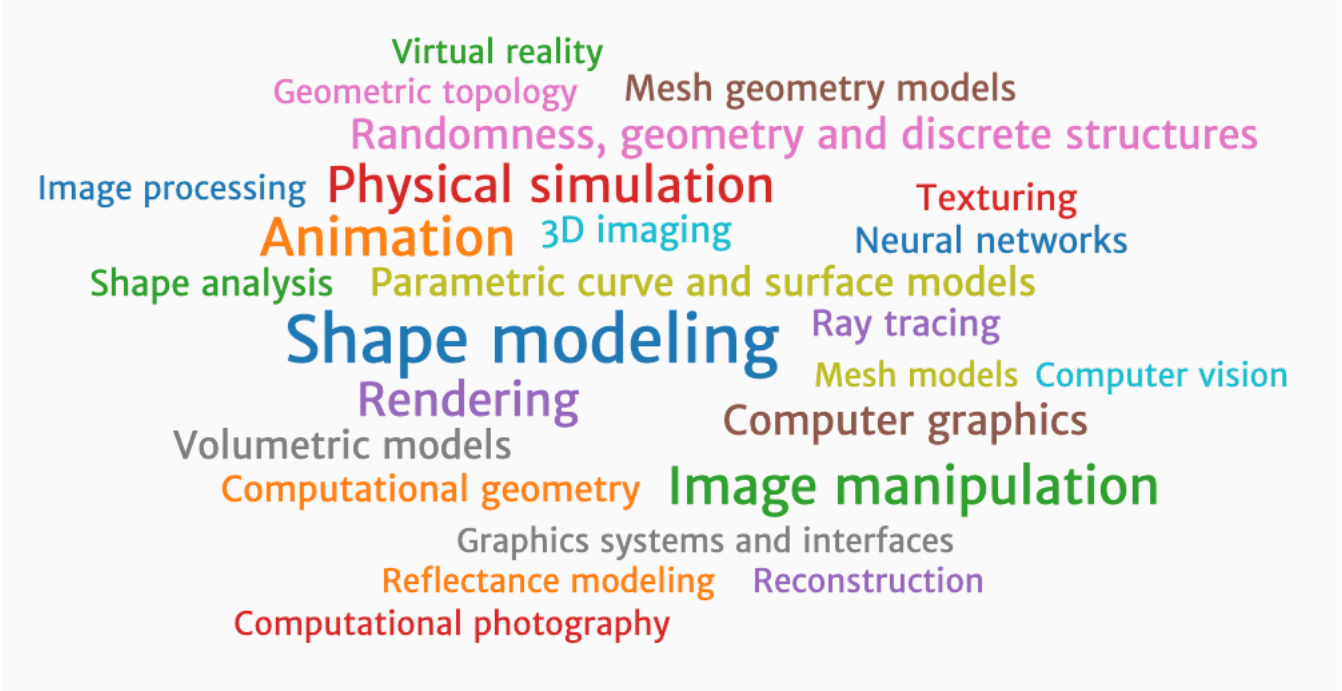
Modeling



Reconstruction

A summary from ACM Transactions on Graphics

"…In the colorful pages of TOG, leading researchers discuss breakthroughs in **computer-aided design, synthetic image generation, rendering, solid modeling, and other areas…**"



Talk outline

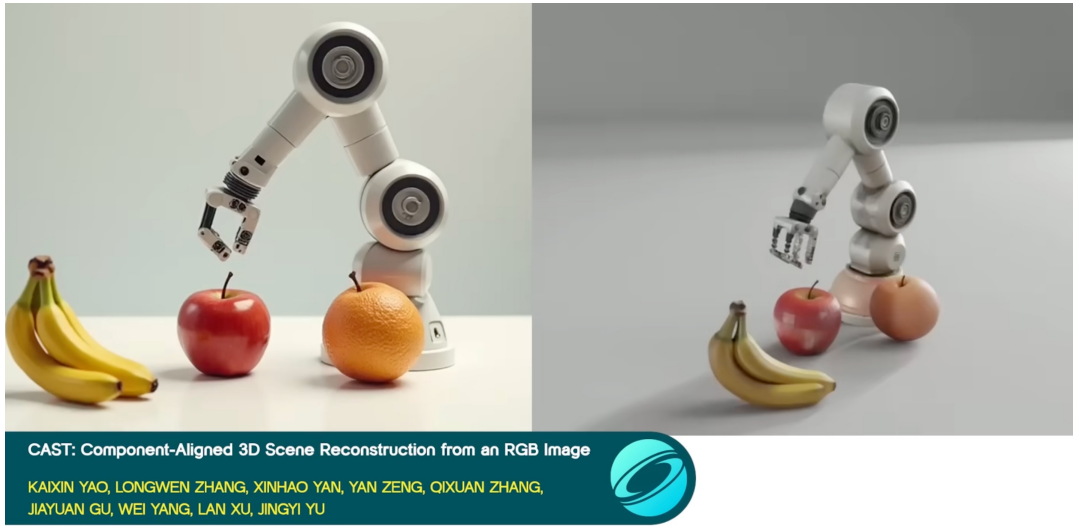
The foundation: building blocks of a virtual world

The next horizon: research frontiers of graphics

SIGGRAPH 2025 technical papers trailer

https://youtu.be/HfHC0wNYry8

The scope of graphics has been much broader than before



CAST: Component-Aligned 3D Scene Reconstruction from an RGB Image

KAIXIN YAO, LONGWEN ZHANG, XINHAO YAN, YAN ZENG, QIXUAN ZHANG, JIAYUAN GU, WEI YANG, LAN XU, JINGYI YU



Cirrus: Adaptive Hybrid Particle-Grid Flow Maps on GPU

MENGDI WANG, FAN FENG, JUNLIN LI, BO ZHU



VR-Doh: Hands-on 3D Modeling in Virtual Reality

ZHAOFENG LUO, ZHITONG CUI, SHIJIAN LUO, MENGYU CHU, MINCHEN LI



PARC: Physics-based Augmentation with Reinforcement Learning for Character Controllers

MICHAEL XU, YI SHI, KANGKANG YIN, XUE BIN PENG

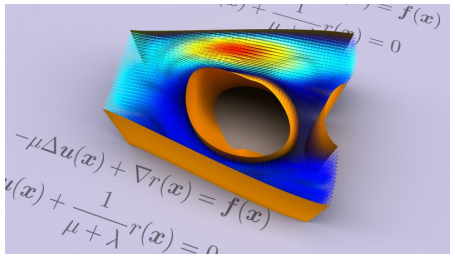
Computational design (2015-2025): an interdisciplinary area

We develop computational tools for designing physical artifacts.

Computational tools

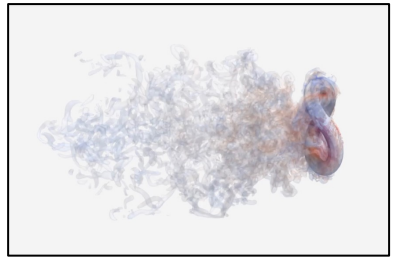
Physical artifacts

Differentiable simulation

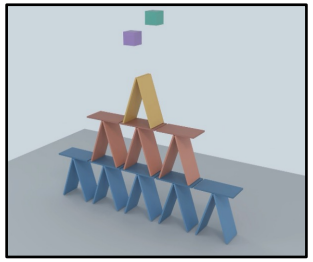




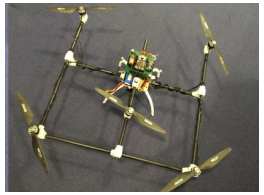
Fluid simulation



LLMs

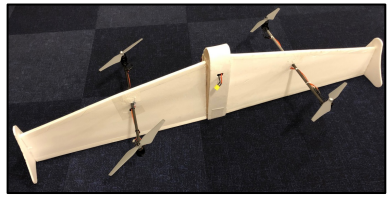


2015

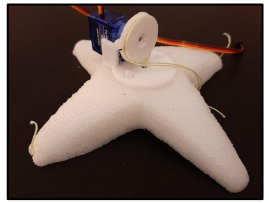


Multirotors and fixed wings

2018

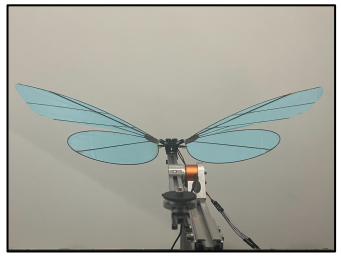


2022



Soft underwater robots

2025

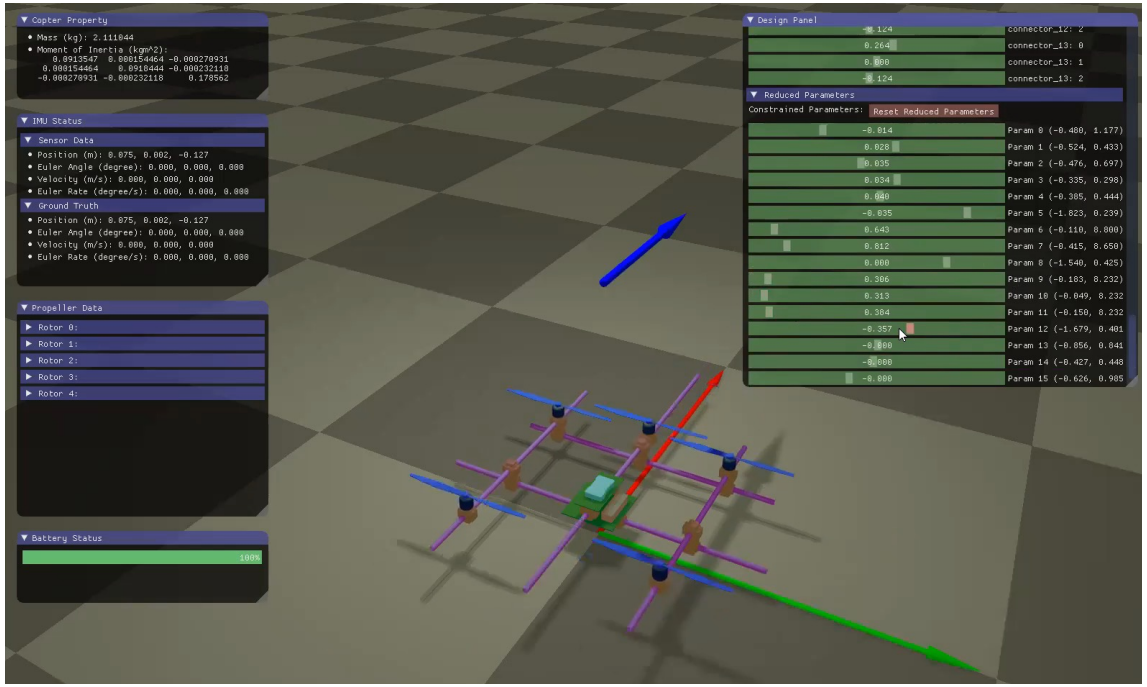


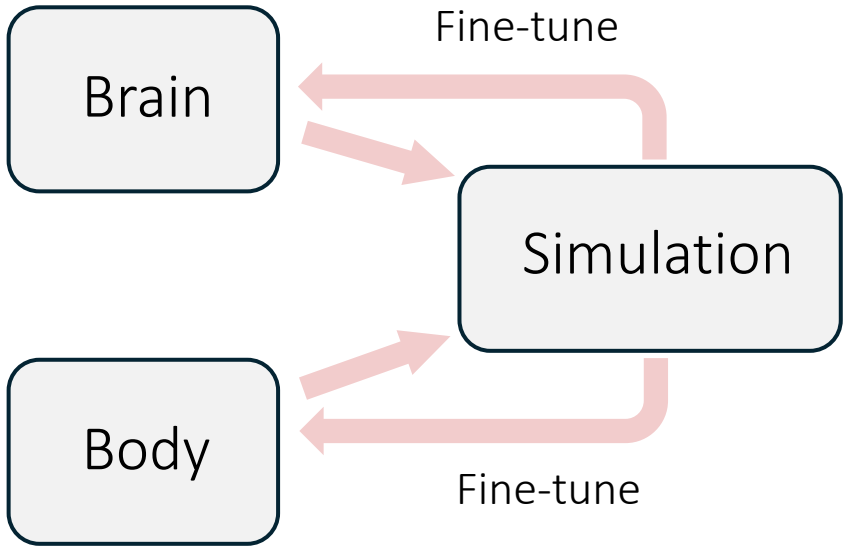
Flapping wings

2026

Rigid robots (2015-2019)

Consider a continuous space of unconventional multicopters.



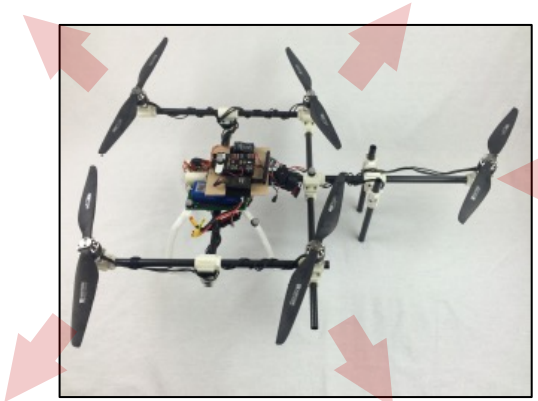


Source: Computational Multicopter Design. ACM TOG 2016.

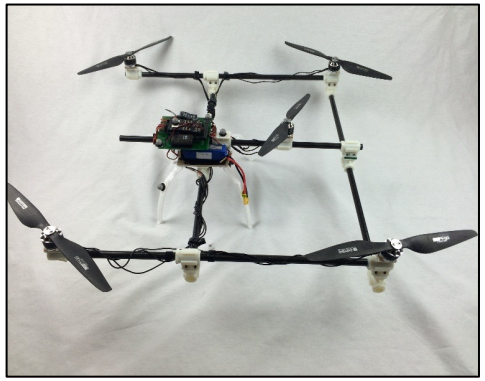
Rigid robots (2015-2019)

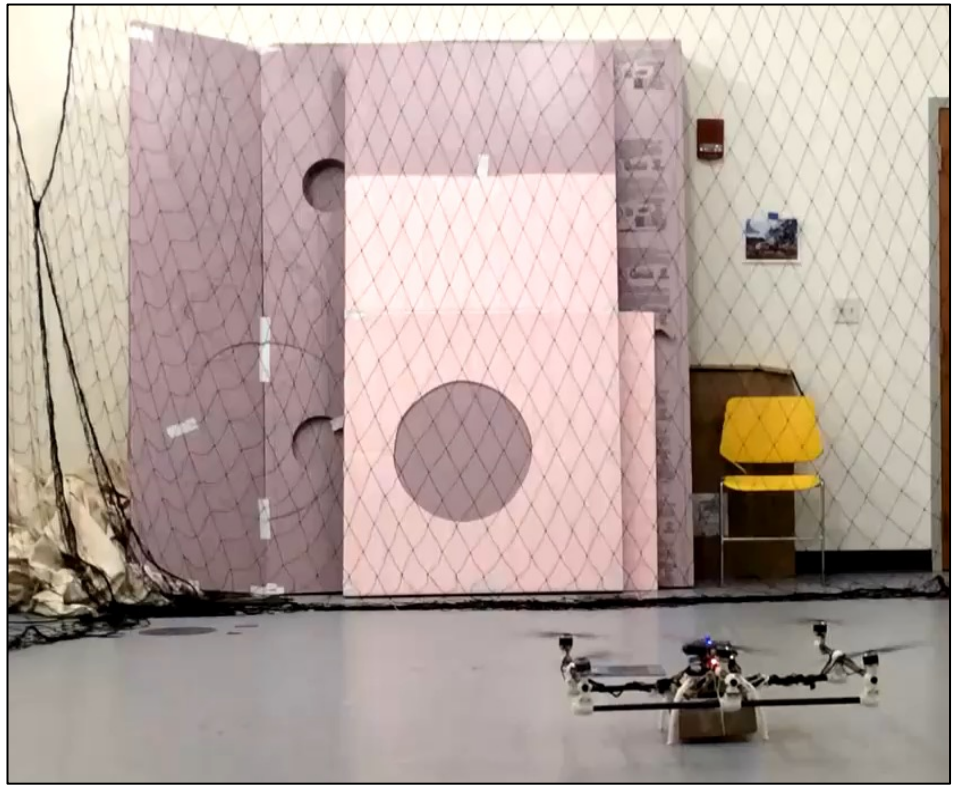
Flight tests (shape):

Old payload:
1047g



New payload:
1392g

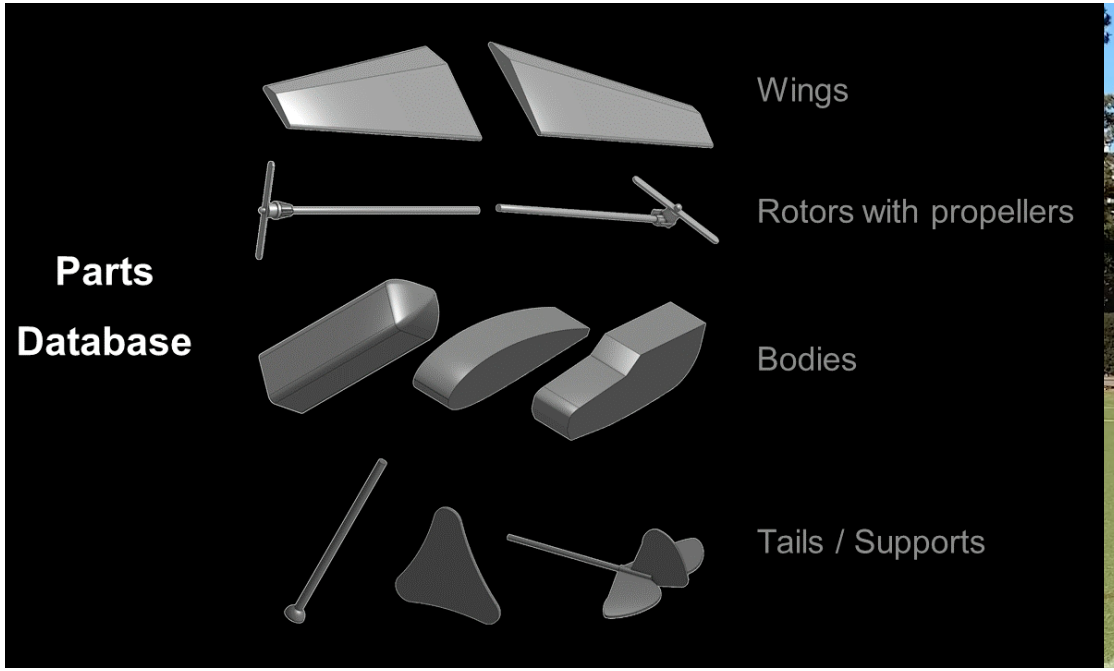


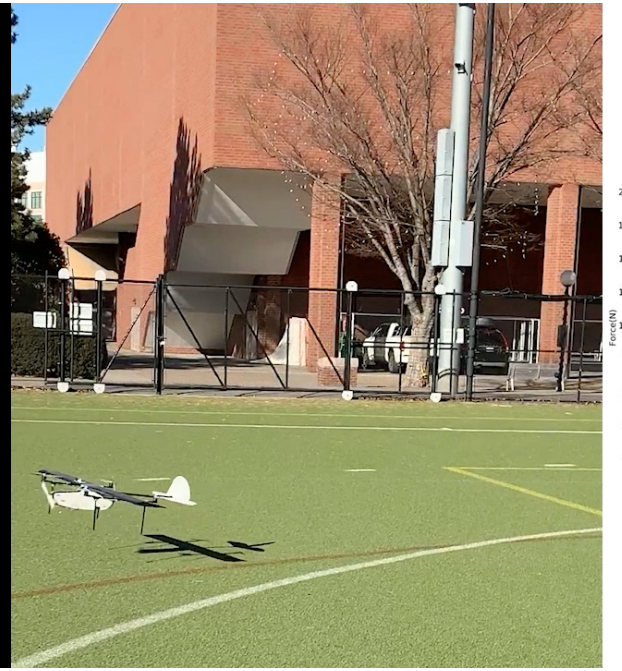


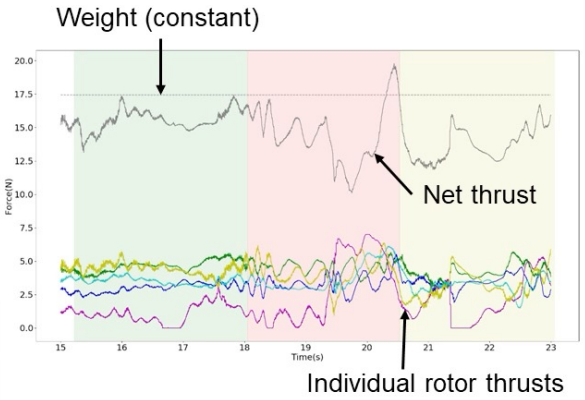
Source: Computational Multicopter Design. ACM TOG 2016.

Rigid robots (2015-2019)

We further extended computational methods to hybrid aerial vehicles.



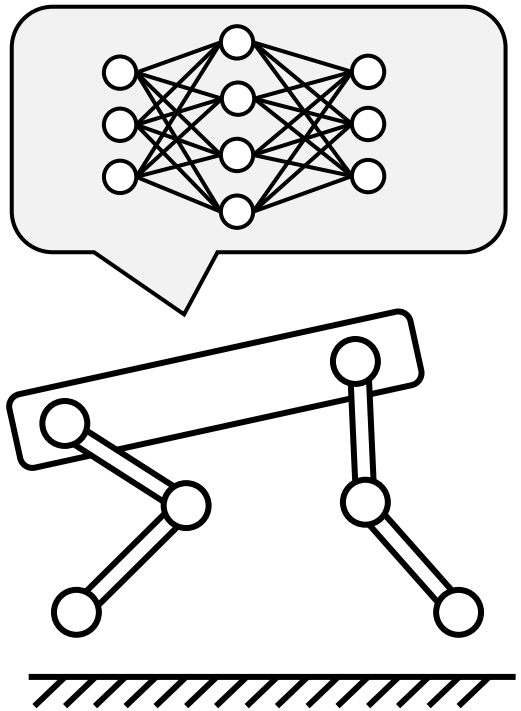




Source: ICLR 2022, IROS 2022, and ACM TOG 2019.

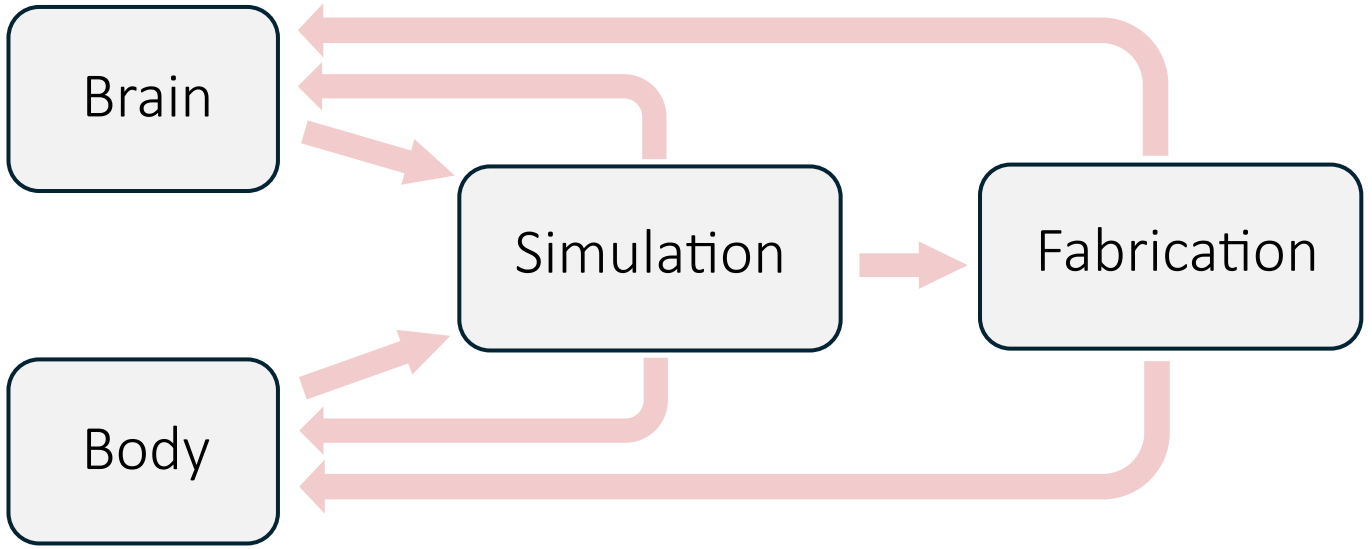
Differentiable simulation (2019-2022)

Design the brain



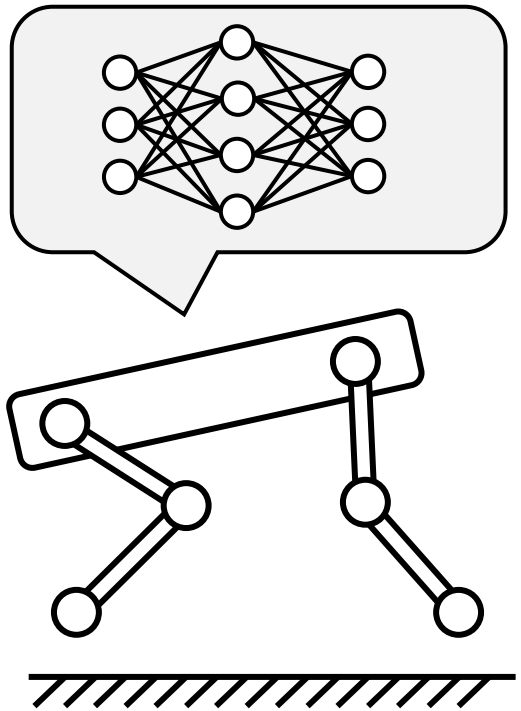
Design the body

A design pipeline:



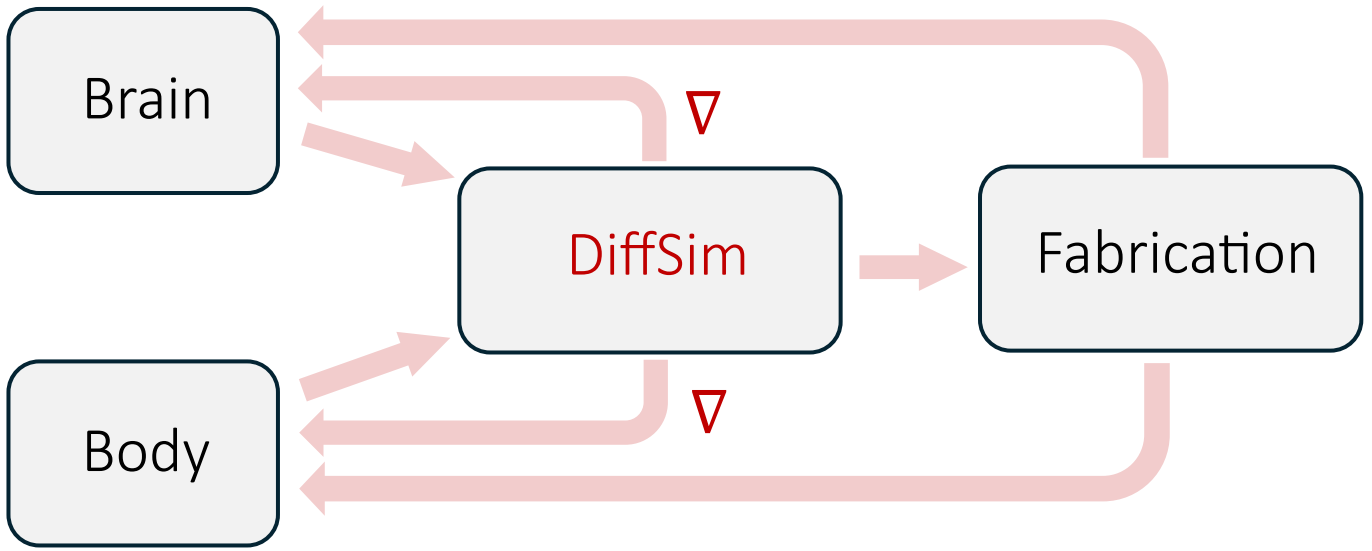
Differentiable simulation (2019-2022)

Design the brain



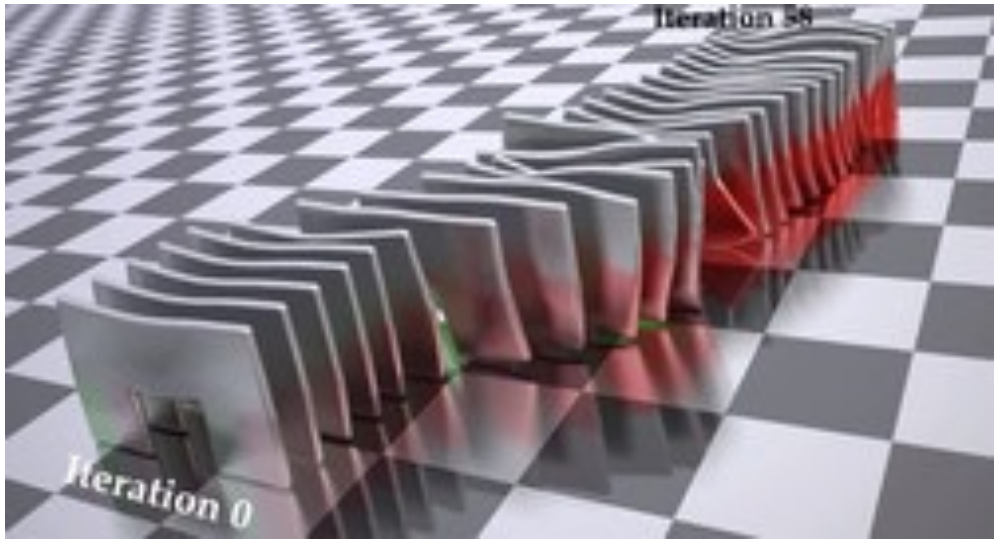
Design the body

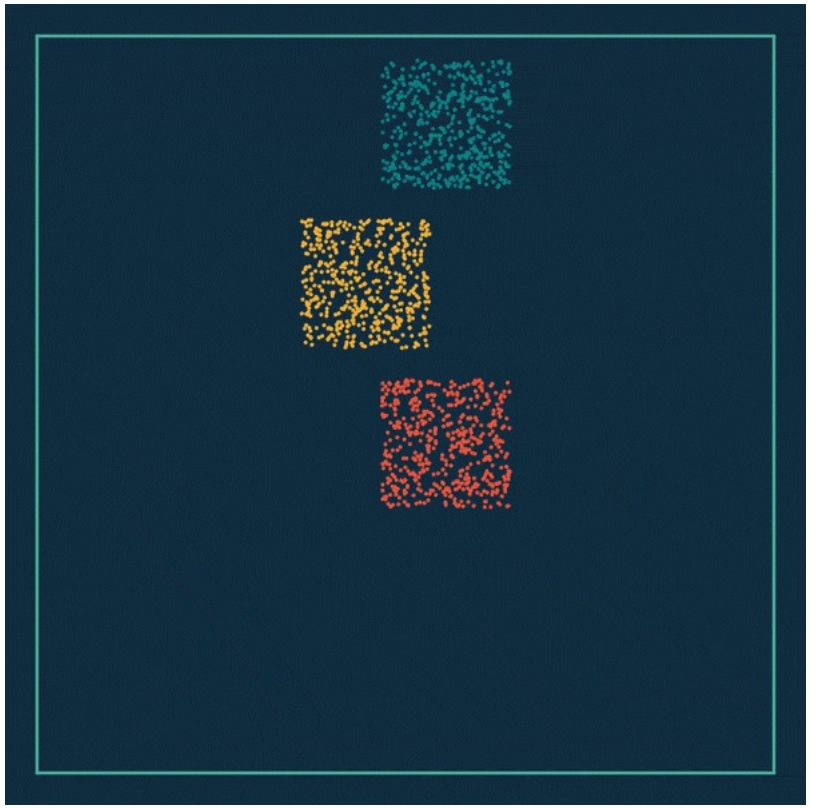
A design pipeline:



Differentiable simulation (2019-2022)

Yuanming Hu developed ChainQueen, a differentiable soft-body simulator in 2019. It later became the famous Taichi Language.

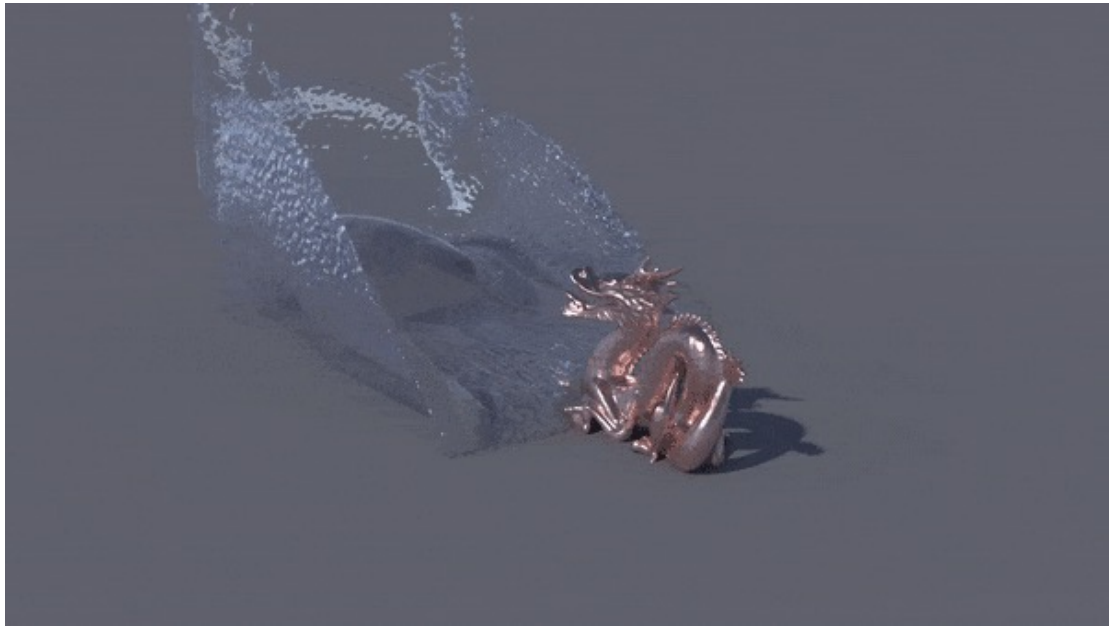




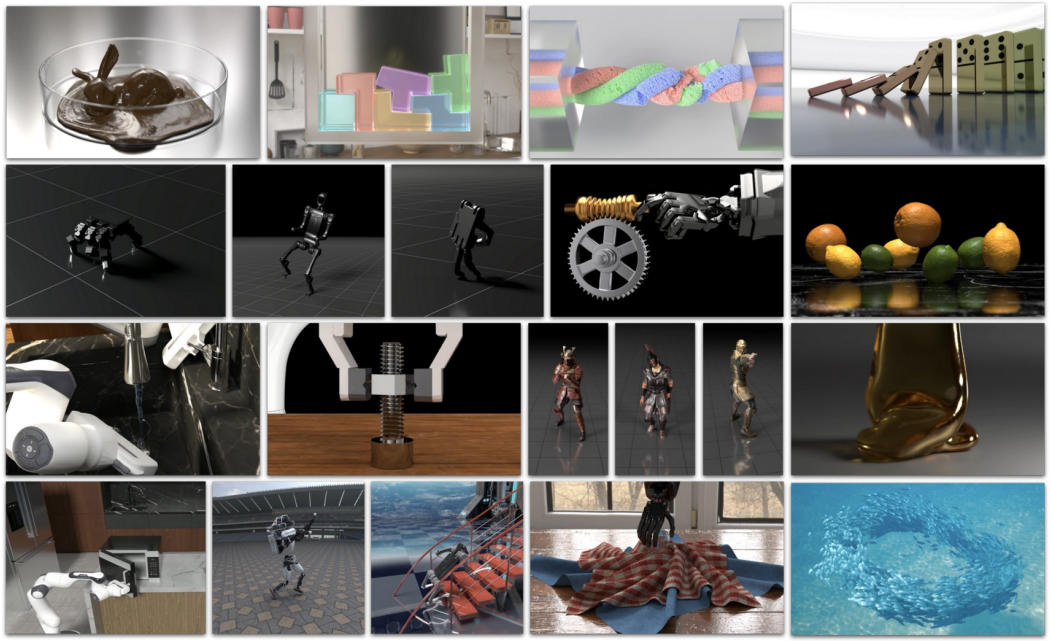
Source: ChainQueen (ICRA 2019) and
https://github.com/yuanming-hu/taichi_mpm

Differentiable simulation (2019-2022)

Differentiable simulation today: Taichi Lang, Nvidia Warp, and Genesis.



https://github.com/erizmr/SPH_Taichi

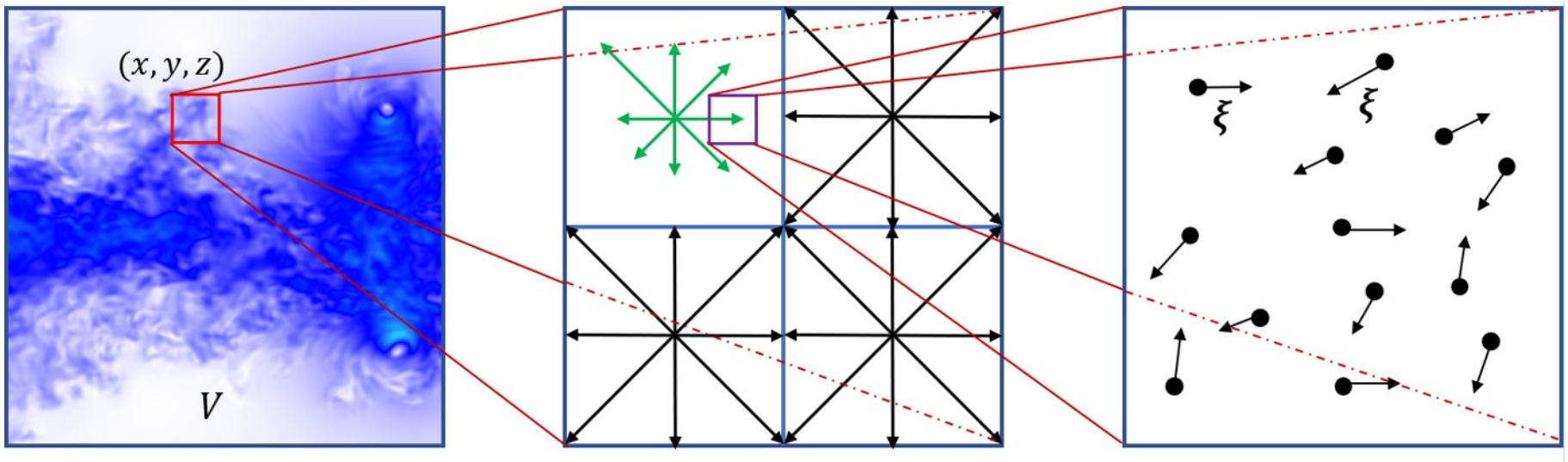


https://github.com/Genesis-Embodied-AI/Genesis

Fluid simulation (2022-)



How to describe flow



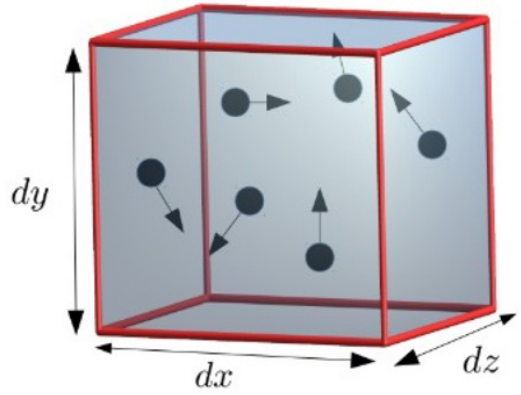
Macroscopic
Navier-Stokes (N-S) equations

Mesoscopic
Boltzmann transport equation

Microscopic
Molecular dynamics

Mesoscopic description

Particle distribution function $f(\boldsymbol{x}, \boldsymbol{v}, t)$

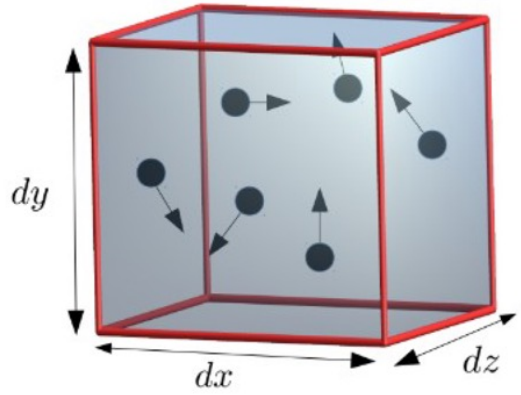


Macroscopic quantities (moments):

Density	$\rho = \int f d\boldsymbol{v}$	Zero order moment
Velocity	$\rho \boldsymbol{u} = \int \boldsymbol{v} f d\boldsymbol{v}$	First order moment
Pressure	$p = \frac{1}{3} \int \lVert \boldsymbol{v} - \boldsymbol{u} \rVert_2^2 f d\boldsymbol{v}$	

Mesoscopic description

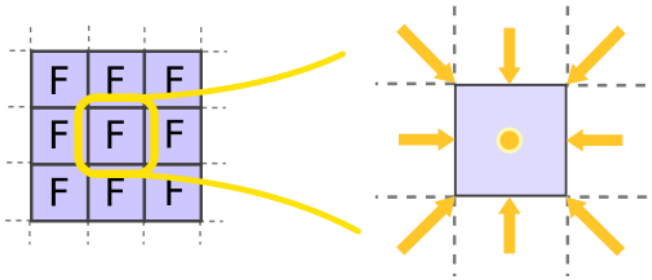
Particle distribution function $f(\boldsymbol{x}, \boldsymbol{v}, t)$



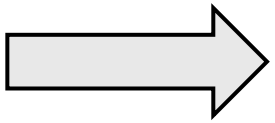
Boltzmann transport equation

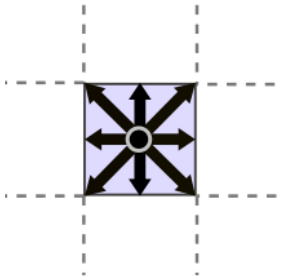
$$\frac{\partial f}{\partial t} + \boldsymbol{v} \cdot \nabla f = \Omega(f) + \boldsymbol{F} \cdot \nabla_{\boldsymbol{v}} f$$

Algorithm overview

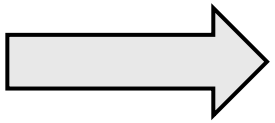


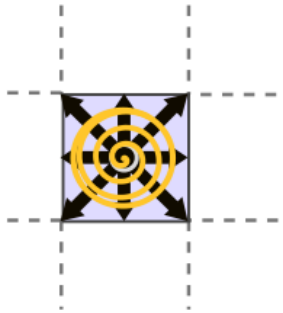
Discretize $f(\boldsymbol{x}, \boldsymbol{v}, t)$



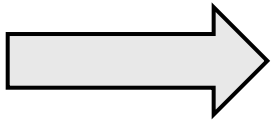


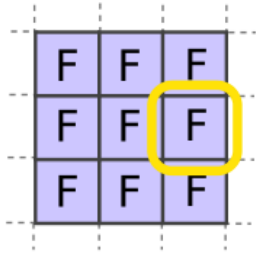
Stream f from adjacent fluid cells





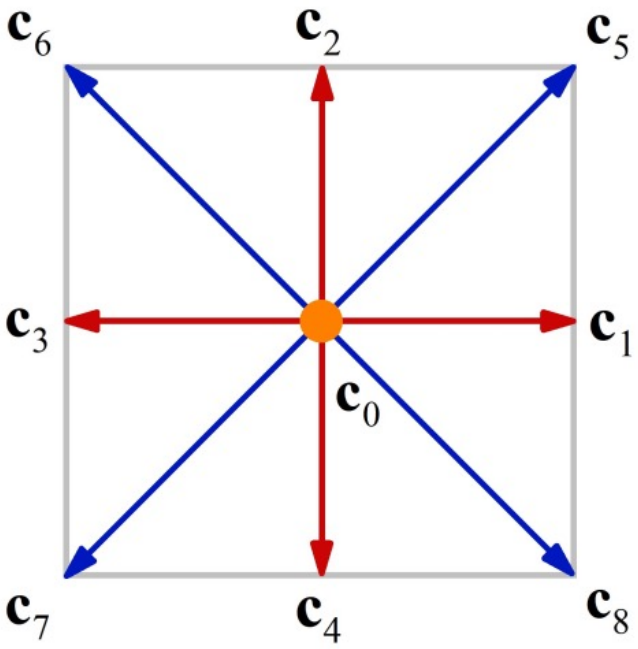
Redistribute f via in-cell collision



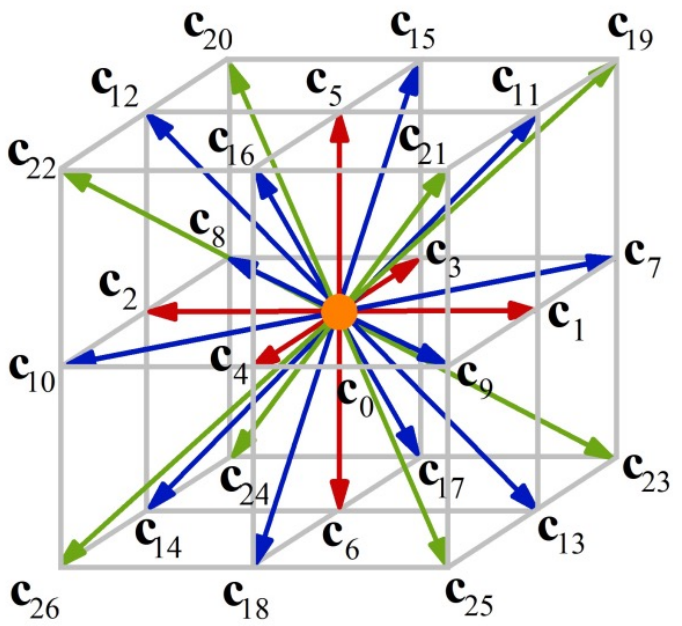


Recover macroscopic properties from f

Multi-speed model

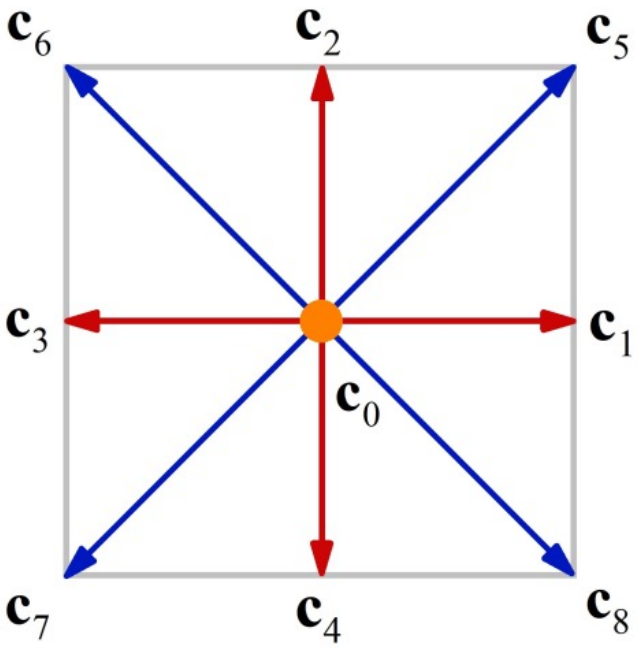


D2Q9 Model



D3Q27 Model

Multi-speed model



D2Q9 Model

$$c_0 = (0,0)$$

$$c_{1,3}, c_{2,4} = (\pm c, 0), (0, \pm c)$$

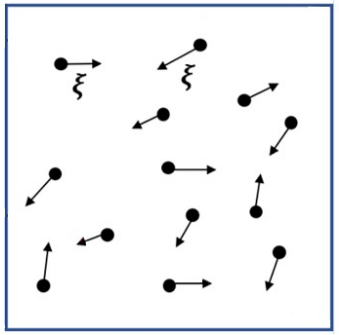
$$c_{5,6,7,8} = (\pm c, \pm c)$$

$$\rho = \int f d\boldsymbol{v} \qquad \rho = \sum_i f_i(\boldsymbol{x}, t)$$

$$\rho \boldsymbol{u} = \int \boldsymbol{v} f d\boldsymbol{v} \qquad \rho \boldsymbol{u} = \sum_i c_i f_i(\boldsymbol{x}, t)$$

Collision

Microscopic view: particles are constantly colliding with one another inside the cell



Mesoscopic view: approaching to the heat equilibrium distribution

$$f^{eq} = \frac{\rho}{(2\pi)^{D/2}} \exp\left(-\frac{\|\boldsymbol{c}-\boldsymbol{u}\|_2^2}{2}\right)$$

Maxwell–Boltzmann distribution

Equilibrium distribution

Taylor expansion of Maxwell–Boltzmann distribution in D2Q9 model

$$f_i^{eq} = \frac{4}{9}\rho\left[1 - \frac{3u^2}{2c^2}\right] \qquad i = 0$$

$$f_i^{eq} = \frac{1}{9}\rho\left[1 + 3\frac{c_i \cdot u}{c^2} + \frac{9}{2}\frac{(c_i \cdot u)^2}{c^4} - \frac{3u^2}{2c^2}\right] \qquad i = 1,2,3,4$$

$$f_i^{eq} = \frac{1}{9}\rho\left[1 + 3\frac{c_i \cdot u}{c^2} + \frac{9}{2}\frac{(c_i \cdot u)^2}{c^4} - \frac{3u^2}{2c^2}\right] \qquad i = 5,6,7,8$$

BJK collision model

Each collision changes the distribution function $f(x, v, t)$ by an amount proportional to the departure of f from equilibrium distribution f^{eq}:

$$\frac{\partial f}{\partial t} = -\frac{f(x, v, t) - f^{eq}(x, v)}{\tau(v)}$$

The relaxation time τ describes how fast the system reaches thermodynamic equilibrium

$$f(x, v, t) - f^{eq}(x, v) = [f(x, v, 0) - f^{eq}(x, v)]e^{-\frac{t}{\tau}}$$
$$f(x, v, t) = (1 - \omega)f(x, v, 0) + \omega f^{eq}(x, v)$$

$$\omega = 1 - e^{-\frac{\Delta t}{\tau}} \approx \frac{1}{\tau} \ (\Delta t = 1 \text{ in LBM})$$

Viscosity

The relaxation time τ is related to the kinematic viscosity ν

$$\nu = c_s^2(\tau - 0.5)$$

LBM simulate fluids of different viscosities by adjusting τ



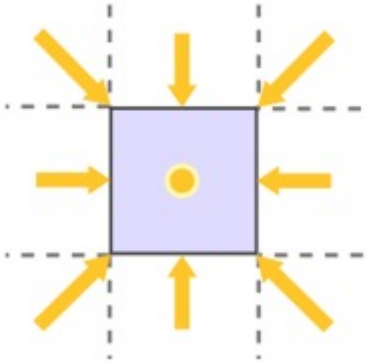


Streaming

Microscopic view: particles are moving to new cells according to their current speed

Mesoscopic view: streaming the distribution to neighbor fluid cells

$$f_i^*(\boldsymbol{x} + \boldsymbol{c}_i, t+1) = f_i(\boldsymbol{x}, t)$$



LBM pseudocode

Initialize:
for all grid nodes (x) do
    Calculate initial density ρ_0 and velocity u_0.
    Compute equilibrium distribution $f_i^{eq}(\rho_0, \mathrm{u}_0)$ for all $i \in [0, 8]$.
    Set current distribution $f_i(\mathrm{x}, t = 0) \leftarrow f_i^{eq}$.
end for

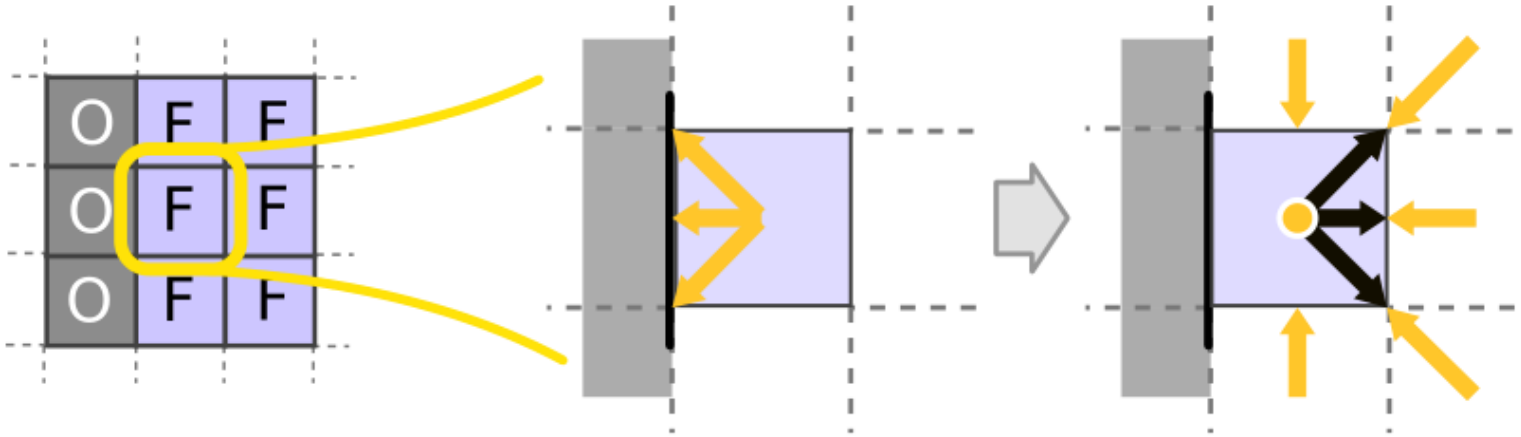
procedure LBM_TimeStep
    for each time step $t \rightarrow t + \Delta t$ do
                ▷ Phase 1: Collision (at each node, fully parallel)
        for all grid nodes (x) do
            1. Compute Macroscopic Quantities:
                $\rho(\mathrm{x}, t) \leftarrow \sum_i f_i(\mathrm{x}, t)$
                $\rho\mathrm{u}(\mathrm{x}, t) \leftarrow \sum_i f_i(\mathrm{x}, t)\mathrm{e}_i$

            2. Compute Equilibrium Distribution:
                Calculate $f_i^{eq}(\rho, \mathrm{u})$ for all i.

            3. Perform Collision (BGK model):
                $f_i^{post}(\mathrm{x}, t) \leftarrow f_i(\mathrm{x}, t) - \frac{1}{\tau}\left(f_i(\mathrm{x}, t) - f_i^{eq}(\rho, \mathrm{u})\right)$
        end for
                ▷ Phase 2: Streaming (fully parallel, ”pull” scheme)
        for all grid nodes (x) do
            for all directions $i \in [0, 8]$ do
                Let source node $\mathrm{x}_{src} \leftarrow \mathrm{x} - \mathrm{e}_i\Delta t$.
                $f_i(\mathrm{x}, t + \Delta t) \leftarrow f_i^{post}(\mathrm{x}_{src}, t)$ ▷ Pull from source node
            end for
        end for
    end for
end procedure

Boundary condition

For no-slip walls:

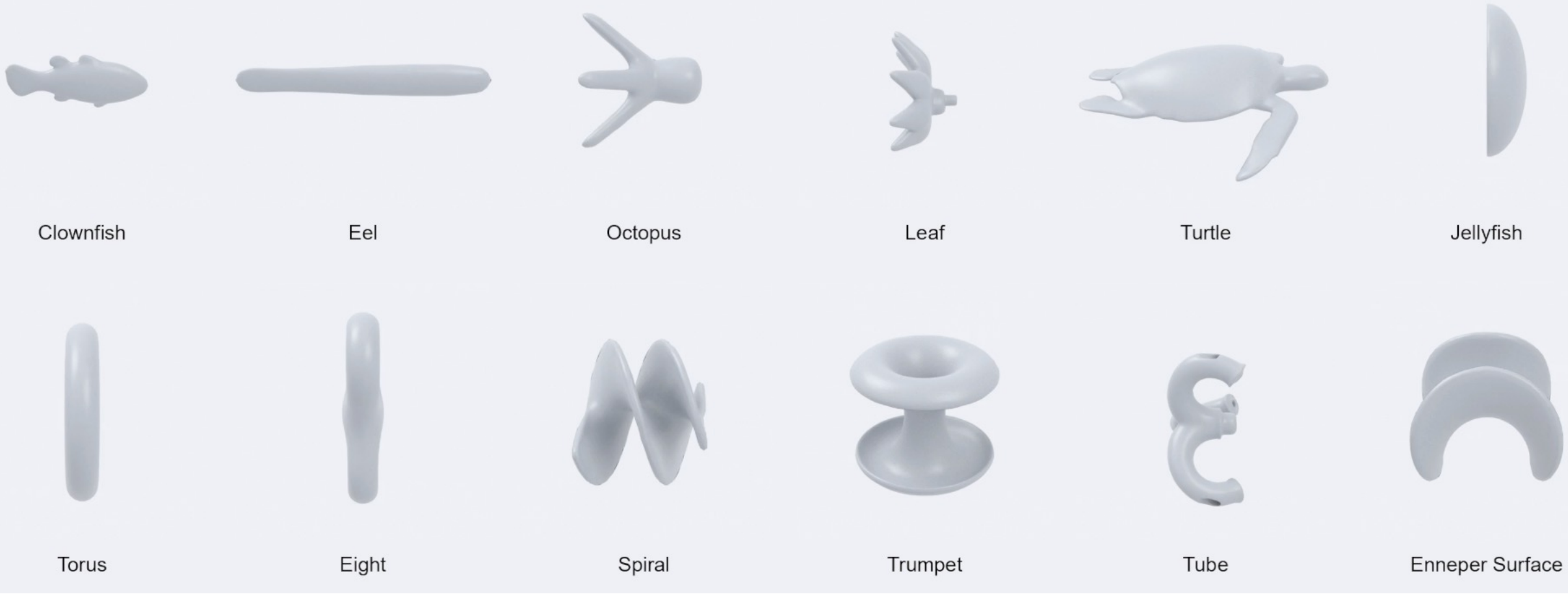


Fluid cell next to obstacles

Reflect f from obstacles

Stream f from adjacent fluid cells

For inlets and outlets: more sophisticated but with many solutions: extrapolation method, Zou-He condition, convective outflow condition, etc.



Learning to control free-form soft swimmers

Changyu Hu, Yanke Qu, Qiuan Yang, Xiaoyu Xiong, Kui Wu, Wei Li, Tao Du

NeurIPS 2025

Physics-based simulation

Physics models

Soft body: hyperelastic materials and finite elements

Fluid: Lattice Boltzmann method (LBM)

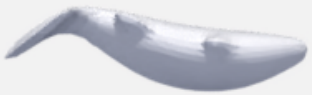
Weak two-way coupling

Performance

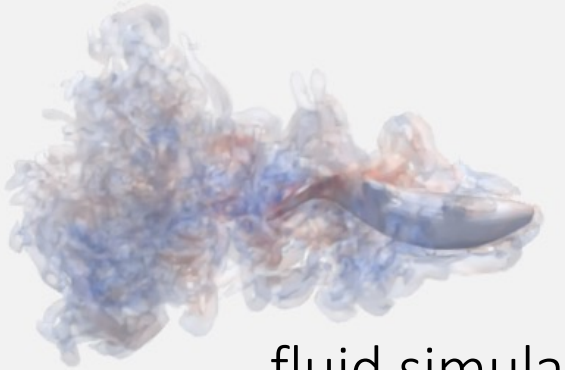
3.5k elements and 128 x 128 x 512 grid resolution

57 fps on one GPU

Source code is available on GitHub



soft-body simulation



fluid simulation

The importance of physics-based fluid for swimming



with physics-based fluid



with simplified fluid

A 2D fish chasing moving targets



It took < 1 hour to train this control policy on six GPUs with soft actor-critic (SAC).

Can a donut swim?



with fluid visualization



without fluid visualization

Can a potato chip swim?

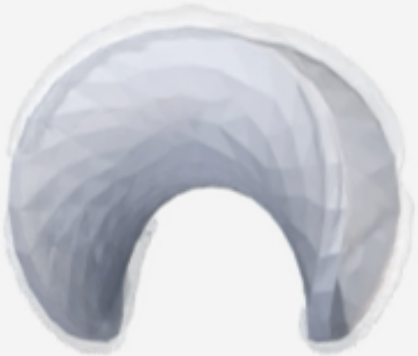


with fluid visualization



without fluid visualization

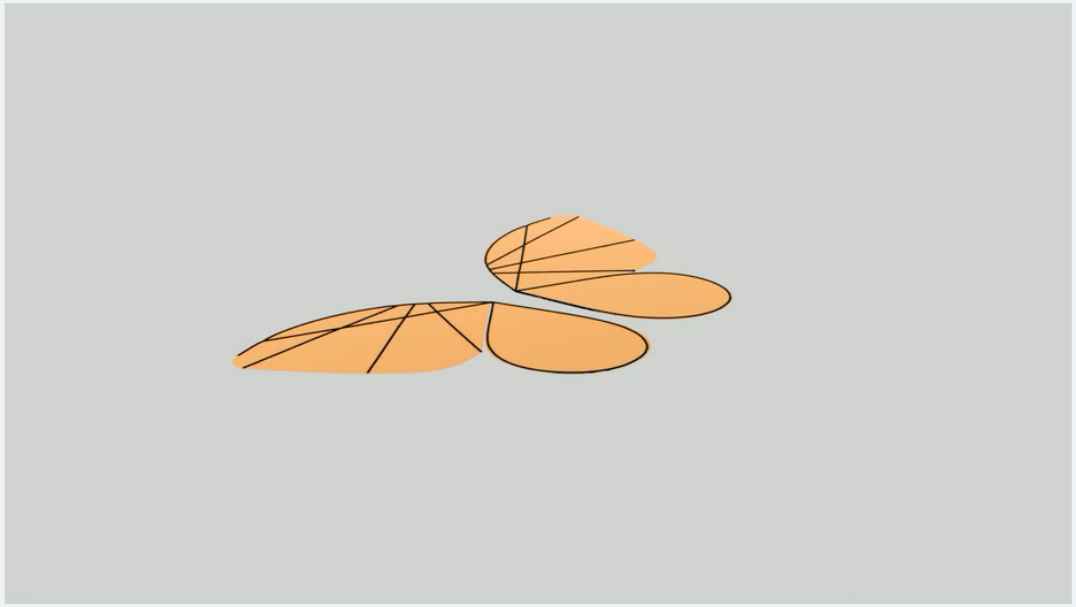
Comparing it to a sea slug





Real-world experiments

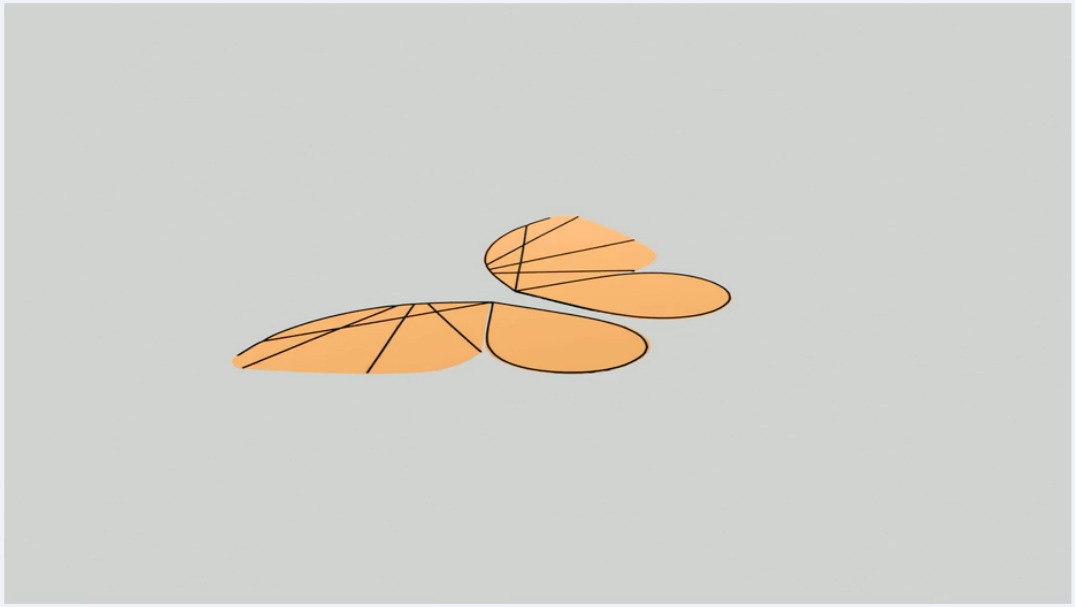
Flow visible



Relative flow speed (qualitative)

Low High

Flow hidden



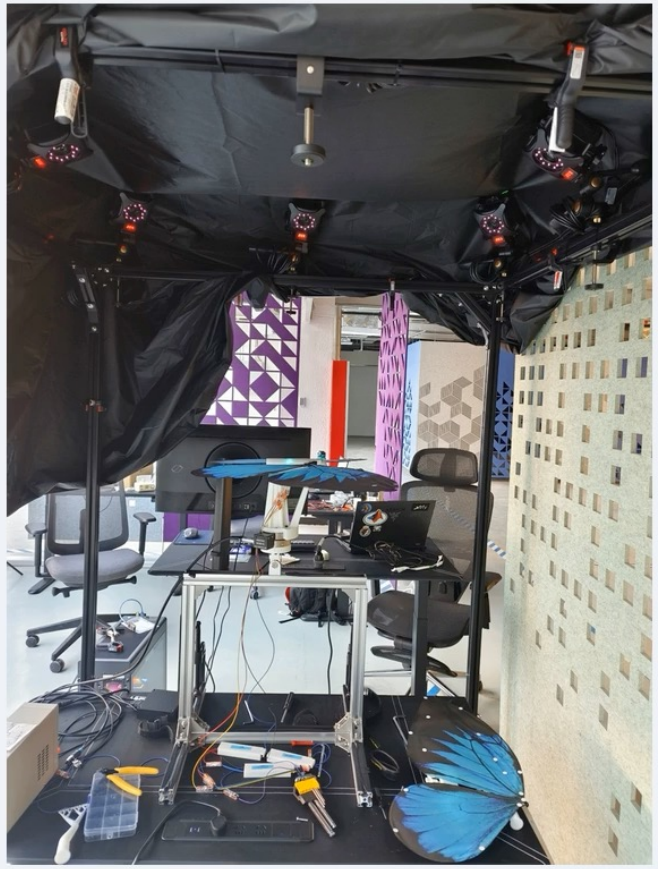
Same motion; only the flow display is hidden

Same simulation, synchronized views · 10 flapping cycles

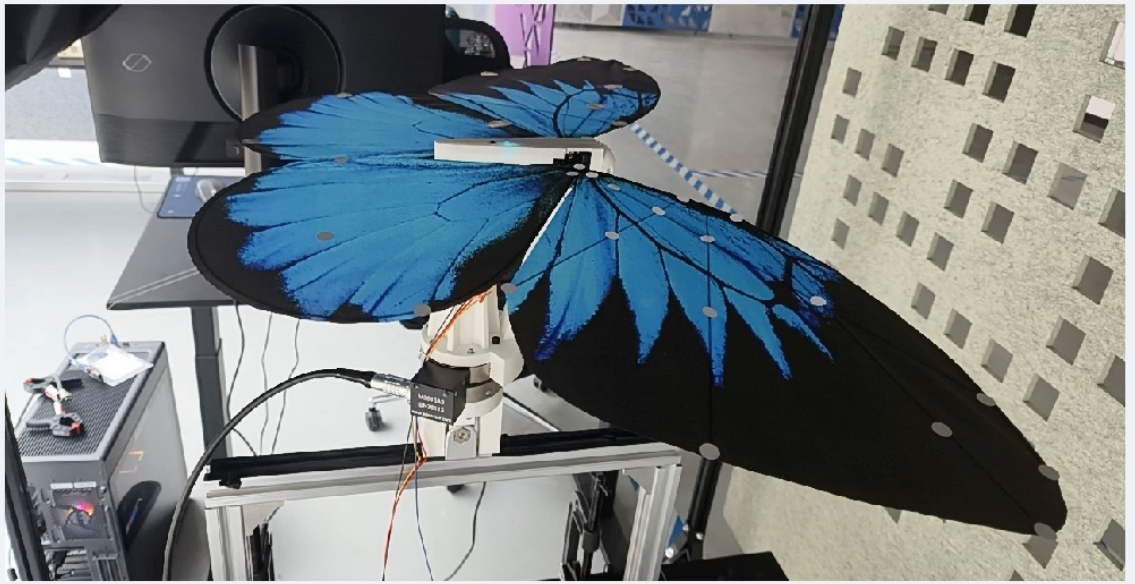
3.509 s physical time · approximately 0.175× playback

Real-world experiments

Flapping-wing measurement platform



Motion-capture cameras and test rig

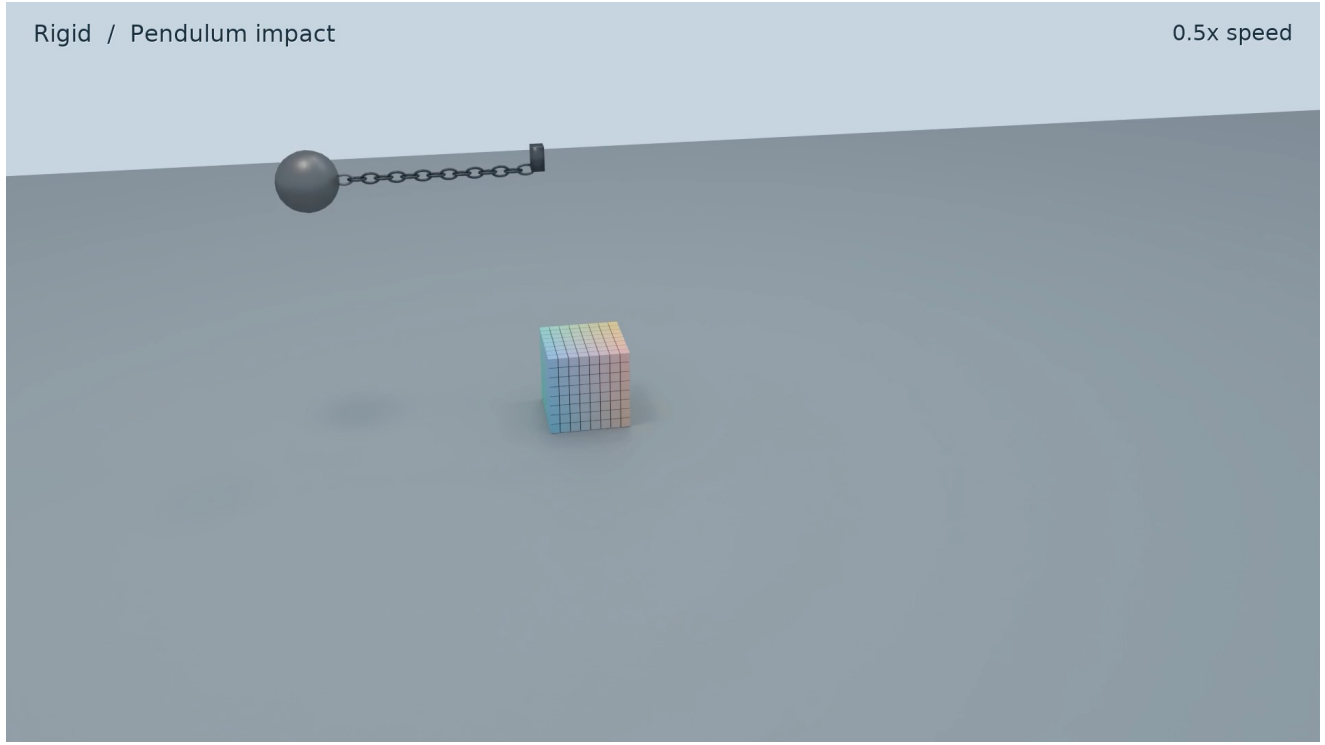


Wing markers and support-force sensing

Large language models and simulation (2026-)

Prompt:

Release a heavy sphere suspended by separate rigid rings into an 8 x 8 x 8 lattice of color cubes, letting the impact scatter the blocks into a settled pile.

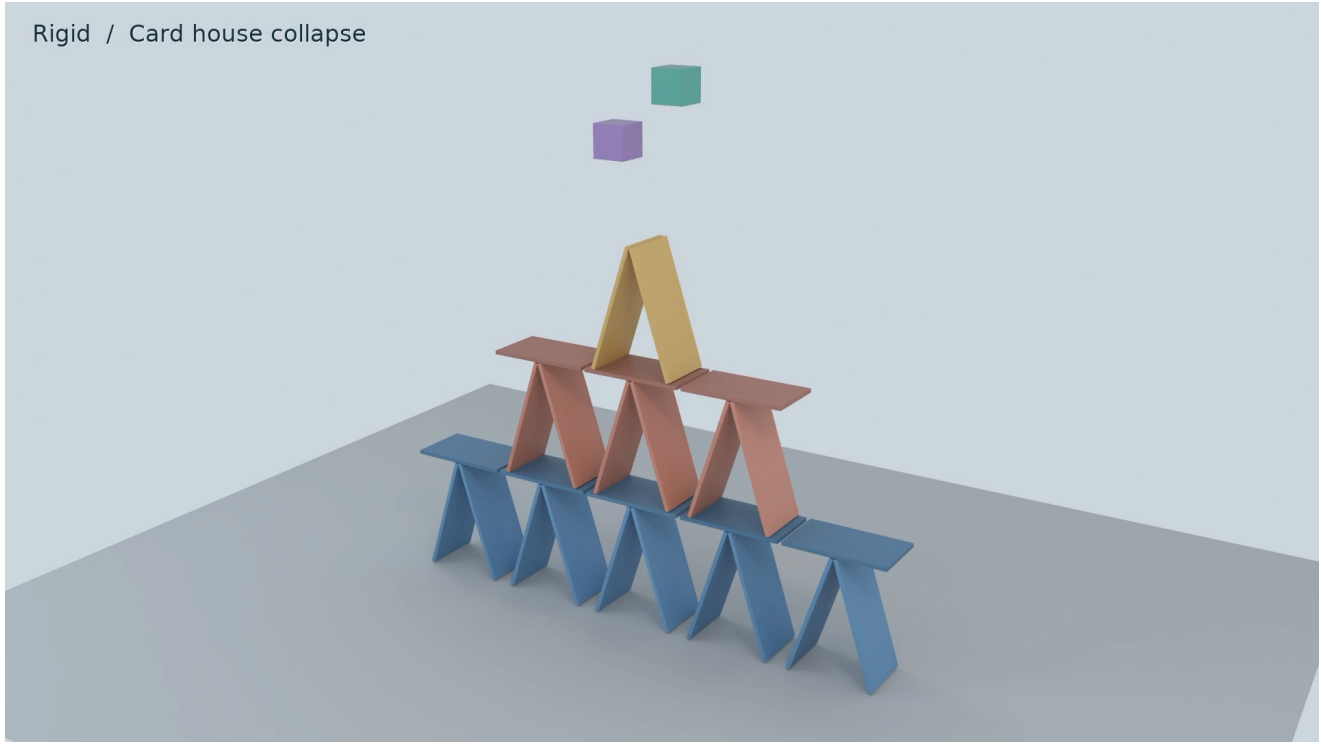


Xiong et al. Text2Sim. arXiv preprint. https://simulation-intelligence.github.io/Text2Sim/

Large language models and simulation (2026-)

Prompt:

Drop two rigid cubes onto a three-level house of cards, with five, three, and one rooms on successive floors, and let it collapse under gravity.



Xiong et al. Text2Sim. arXiv preprint. https://simulation-intelligence.github.io/Text2Sim/

The end

Works in collaboration with my students/colleges:

Xiaoyu Xiong (IIIS undergrad), Changyu Hu, Yanke Qu, Qiuan Yang (IIIS Ph.D. students), Tsun-Hsuan Wang, Yi-Ling Qiao (Genesis AI), and Minchen Li (CMU).

Questions -> taodu@tsinghua.edu.cn

Thank you.